PCB Artwork Design

EOM MIN HO

INDEX

1. **Project Overview**
   - 1.1 Schematic: Decade Counter LED Sequencer
   - 1.2 Parts List & BOM
   - 1.3 Component Footprint Mapping / Cross Reference

2. **Pre-Layout Setup**
   - 2.1 PCB Editor Initial Settings (Design Parameters)
   - 2.2 Board Outline Setup
   - 2.3 Constraints Setup (Net/Physical) — show Rats for VCC/GND

3. **Layout**
   - 3.1 Part Placement and Routing
   - 3.2 Ground Copper Pour Placement
   - 3.3 Color Settings
   - 3.4 Reference Annotation & Uniform Reference Size
   - 3.5 Design Rule Check (DRC)

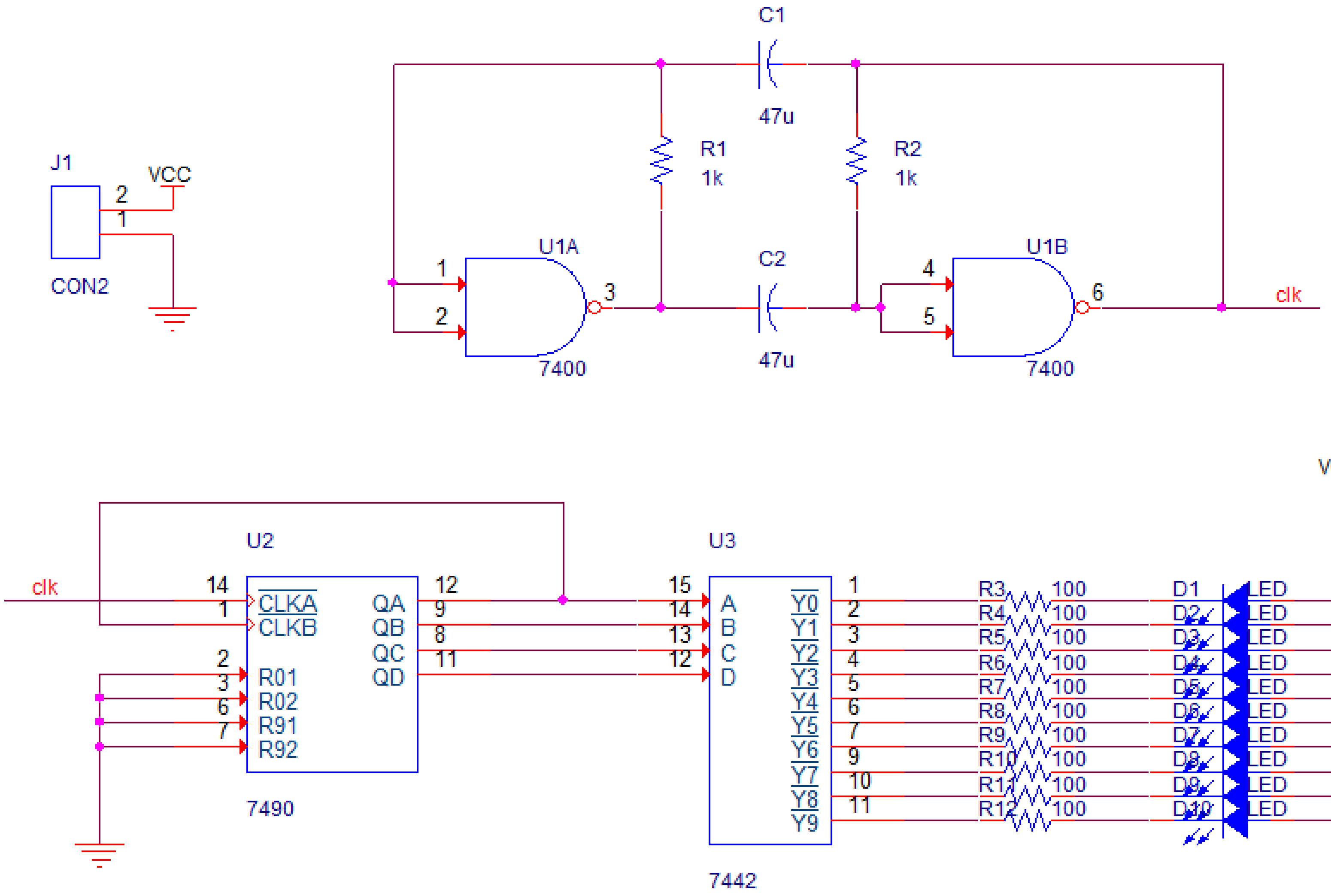
4. **Drafting & Dimensions**
   - 4.1 Drafting Dimension Settings
   - 4.2 Add Dimensions
   - 4.3 Drill Legend Table

5. **Manufacturing Outputs**
   - 5.1 Generate Gerber
   - 5.2 Generate NC Drill
   - 5.3 Gerber Layers (Top Copper, Bottom Copper, Top Soldermask, Top Silkscreen, Drill Drawing)

6. **Contact**

1.Project Overview

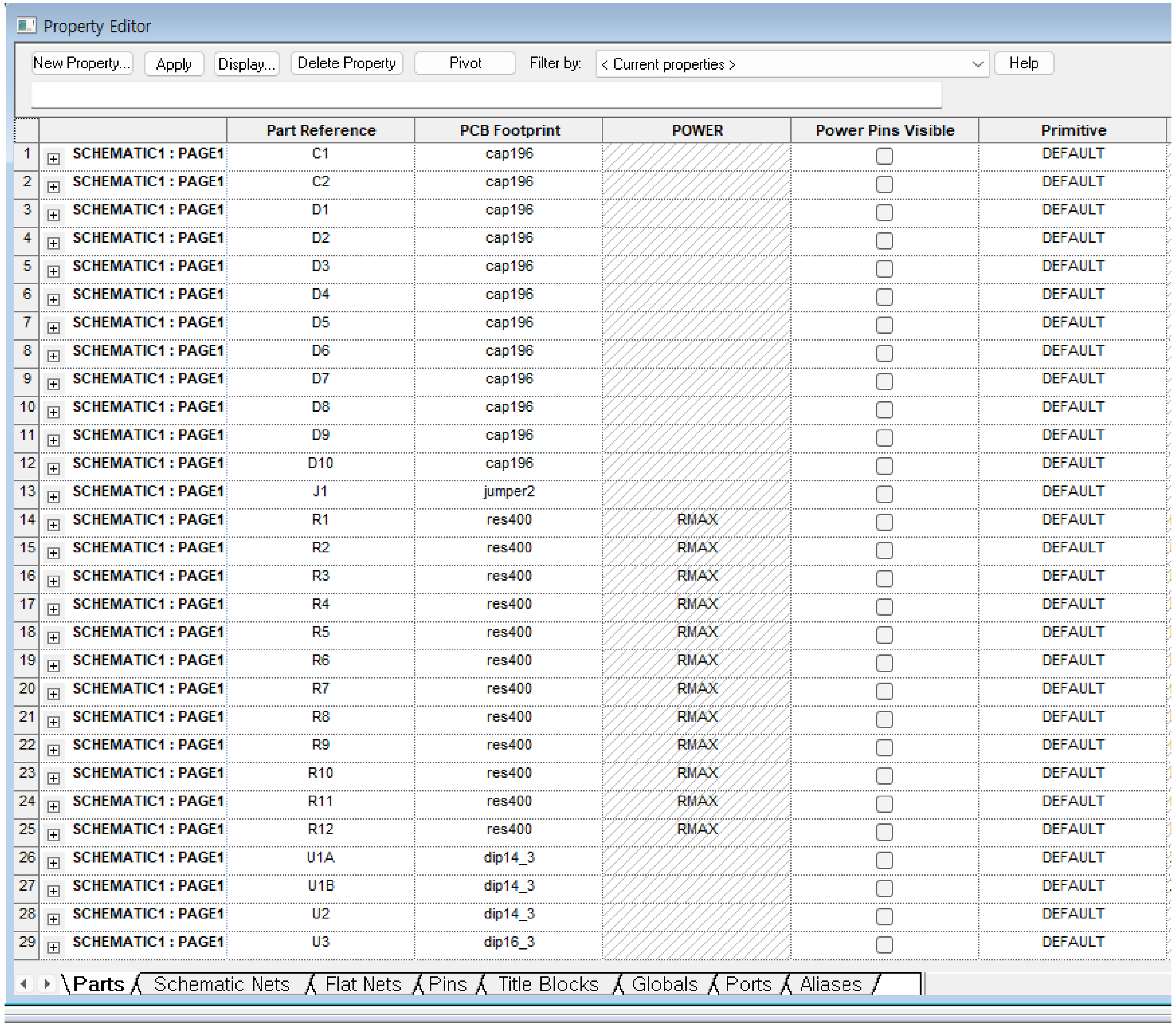
Decade Counter LED Sequencer – Schematic Diagram



Parts List

No.	Part Name (Capture)	Value	Qty	Library
1	CON2	2-pin	1	Connector.olb
2	C	47 μF	2	Discrete.olb
3	R	1 kΩ 100 Ω	2 10	analog.olb
4	7400	NAND Gate	2	eval.olb
5	7490	Decade Counter	1	Counter.olb
6	7442	BCD to Decimal Decoder	1	MuxDecoder.olb
7	D1–D10	LED (Red)	10	Discrete.olb

Component Footprint



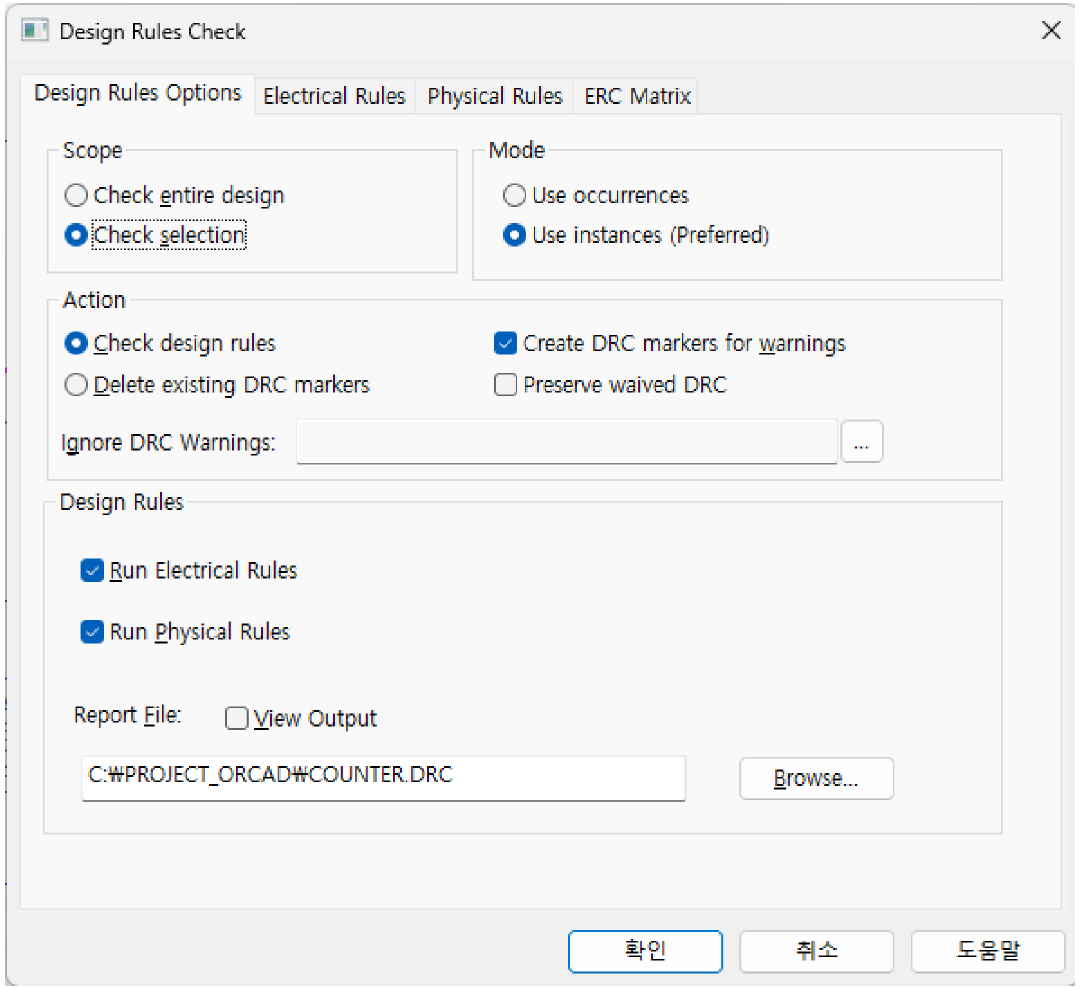
Property Editor

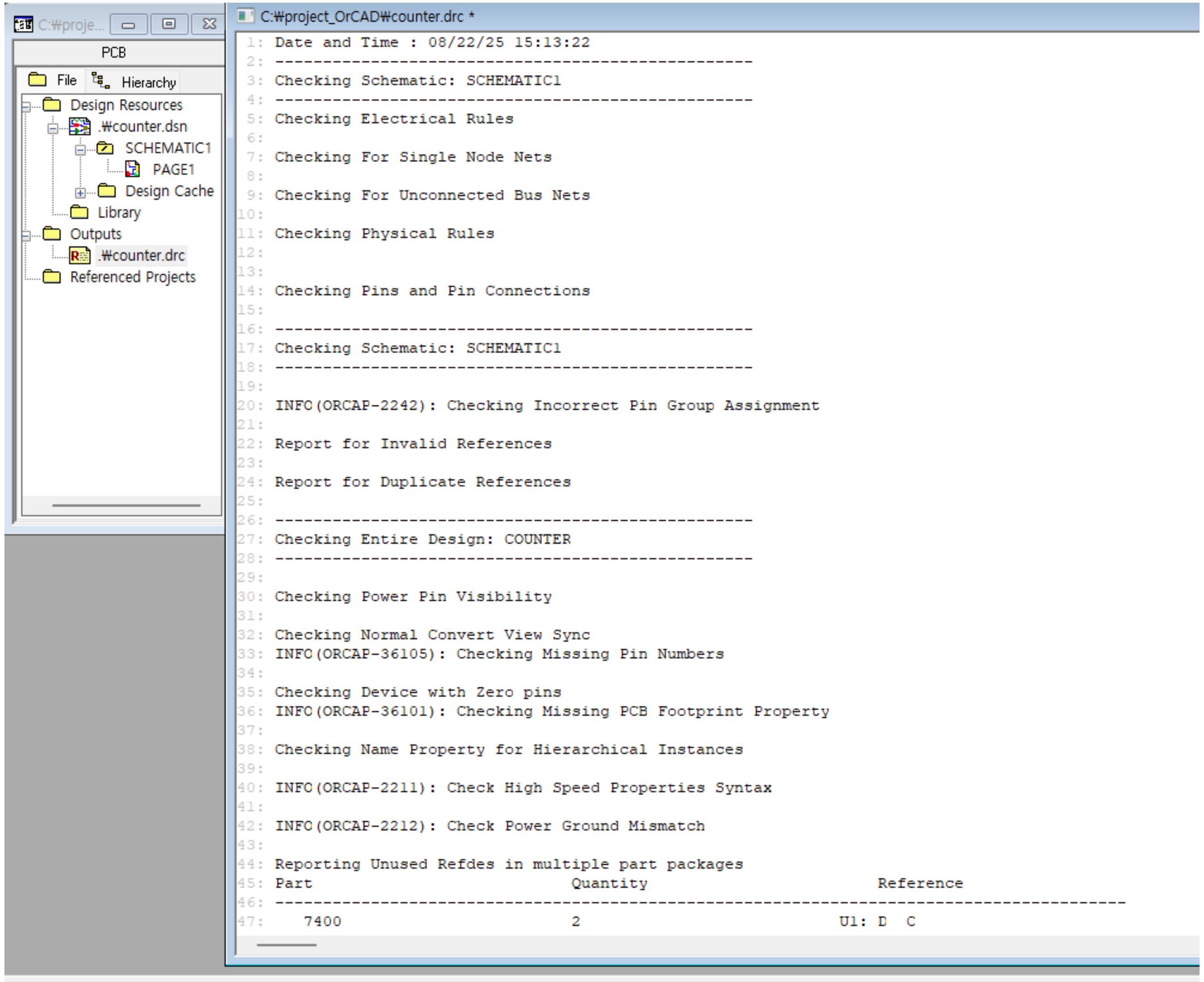
New Property... Apply Display... Delete Property Pivot Filter by: < Current properties > Help

		Part Reference	PCB Footprint	POWER	Power Pins Visible	Primitive
1	SCHEMATIC1 : PAGE1	C1	cap196			DEFAULT
2	SCHEMATIC1 : PAGE1	C2	cap196			DEFAULT
3	SCHEMATIC1 : PAGE1	D1	cap196			DEFAULT
4	SCHEMATIC1 : PAGE1	D2	cap196			DEFAULT
5	SCHEMATIC1 : PAGE1	D3	cap196			DEFAULT
6	SCHEMATIC1 : PAGE1	D4	cap196			DEFAULT
7	SCHEMATIC1 : PAGE1	D5	cap196			DEFAULT
8	SCHEMATIC1 : PAGE1	D6	cap196			DEFAULT
9	SCHEMATIC1 : PAGE1	D7	cap196			DEFAULT
10	SCHEMATIC1 : PAGE1	D8	cap196			DEFAULT
11	SCHEMATIC1 : PAGE1	D9	cap196			DEFAULT
12	SCHEMATIC1 : PAGE1	D10	cap196			DEFAULT
13	SCHEMATIC1 : PAGE1	J1	jumper2			DEFAULT
14	SCHEMATIC1 : PAGE1	R1	res400	RMAX		DEFAULT
15	SCHEMATIC1 : PAGE1	R2	res400	RMAX		DEFAULT
16	SCHEMATIC1 : PAGE1	R3	res400	RMAX		DEFAULT
17	SCHEMATIC1 : PAGE1	R4	res400	RMAX		DEFAULT
18	SCHEMATIC1 : PAGE1	R5	res400	RMAX		DEFAULT
19	SCHEMATIC1 : PAGE1	R6	res400	RMAX		DEFAULT
20	SCHEMATIC1 : PAGE1	R7	res400	RMAX		DEFAULT
21	SCHEMATIC1 : PAGE1	R8	res400	RMAX		DEFAULT
22	SCHEMATIC1 : PAGE1	R9	res400	RMAX		DEFAULT
23	SCHEMATIC1 : PAGE1	R10	res400	RMAX		DEFAULT
24	SCHEMATIC1 : PAGE1	R11	res400	RMAX		DEFAULT
25	SCHEMATIC1 : PAGE1	R12	res400	RMAX		DEFAULT
26	SCHEMATIC1 : PAGE1	U1A	dip14_3			DEFAULT
27	SCHEMATIC1 : PAGE1	U1B	dip14_3			DEFAULT
28	SCHEMATIC1 : PAGE1	U2	dip14_3			DEFAULT
29	SCHEMATIC1 : PAGE1	U3	dip16_3			DEFAULT

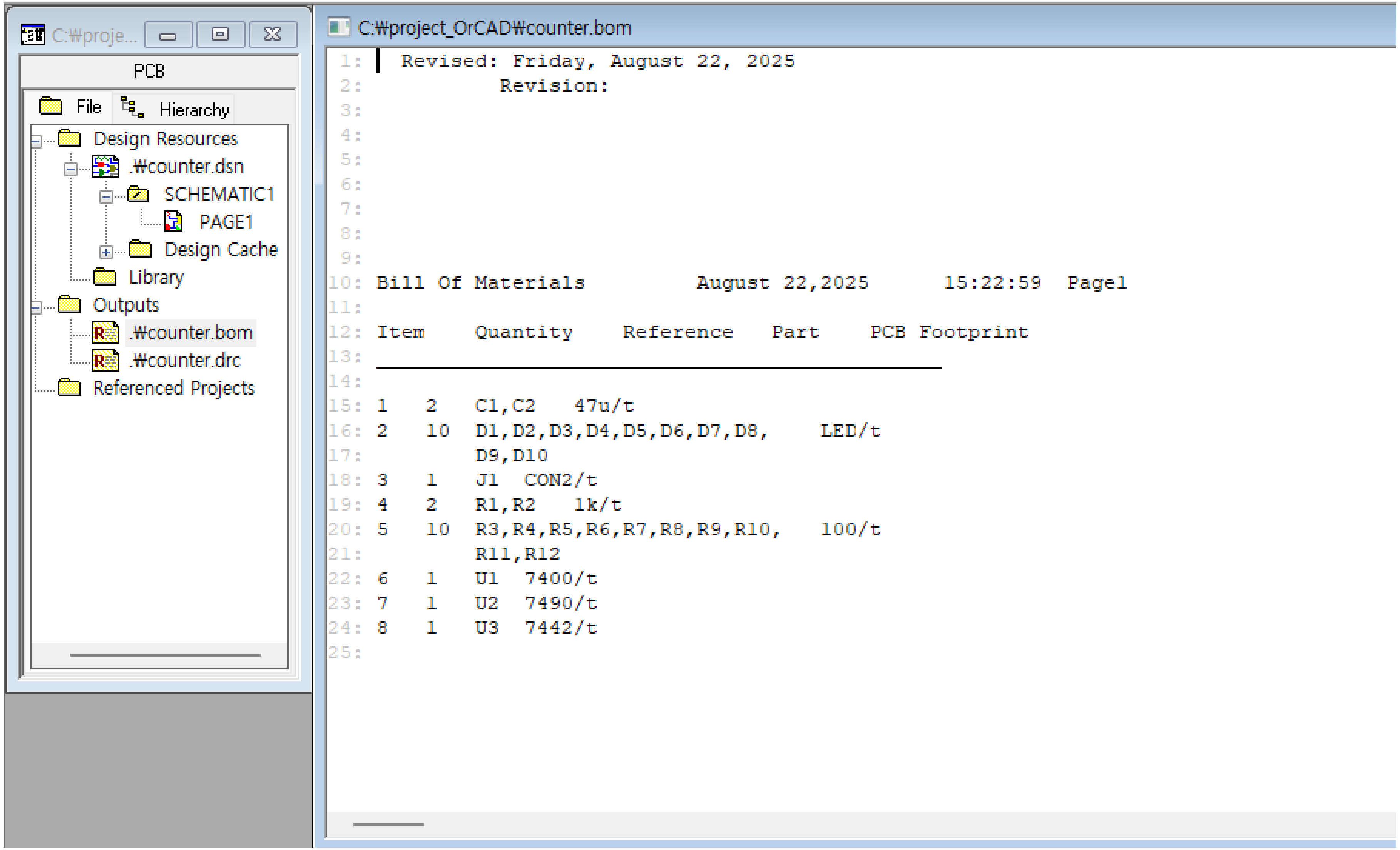
Parts / Schematic Nets / Flat Nets / Pins / Title Blocks / Globals / Ports / Aliases

Design Rule Check (DRC)

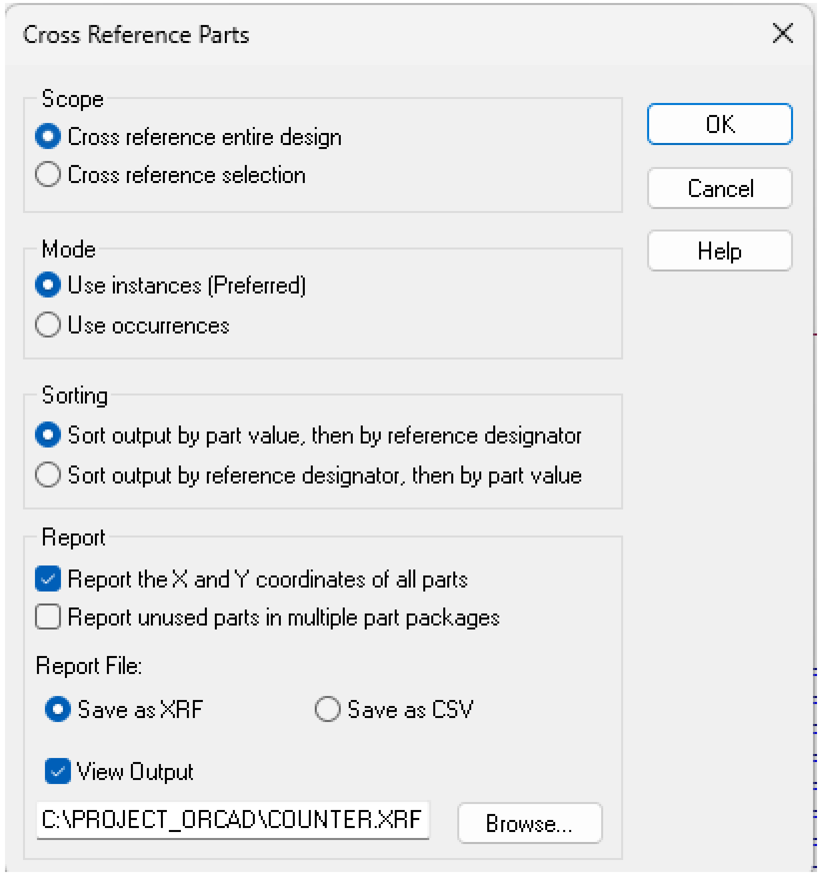


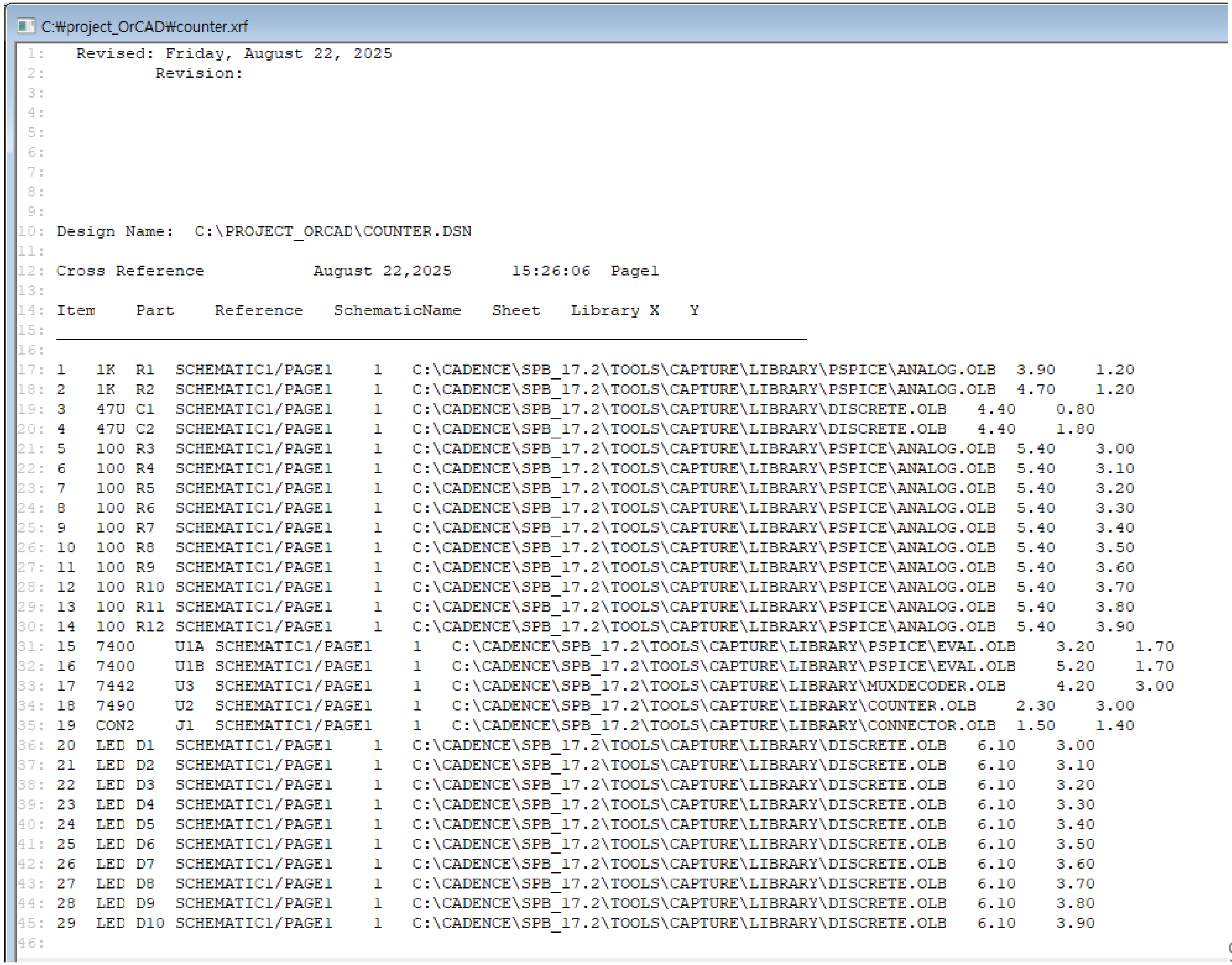


Bill of Materials(BOM)

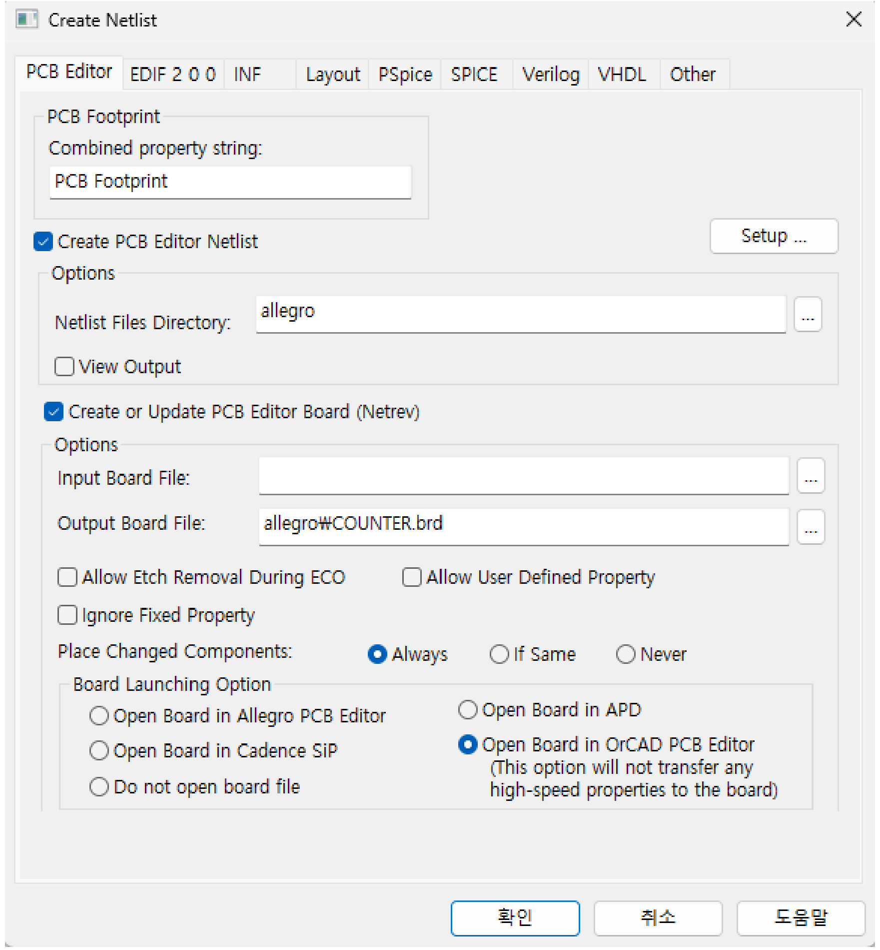


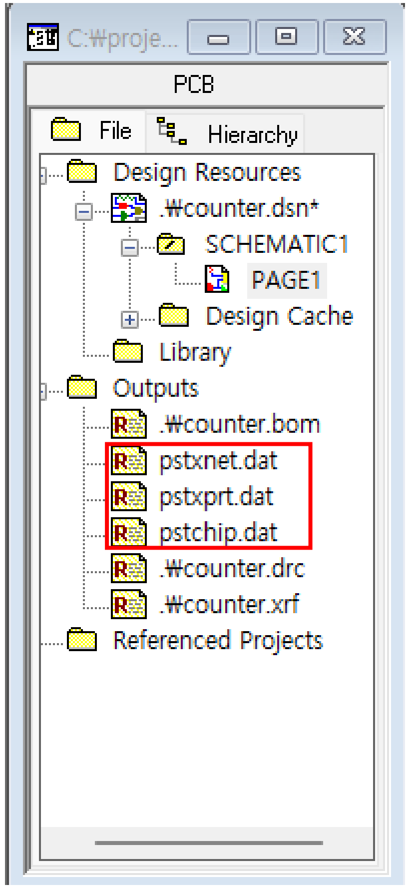
Cross Reference





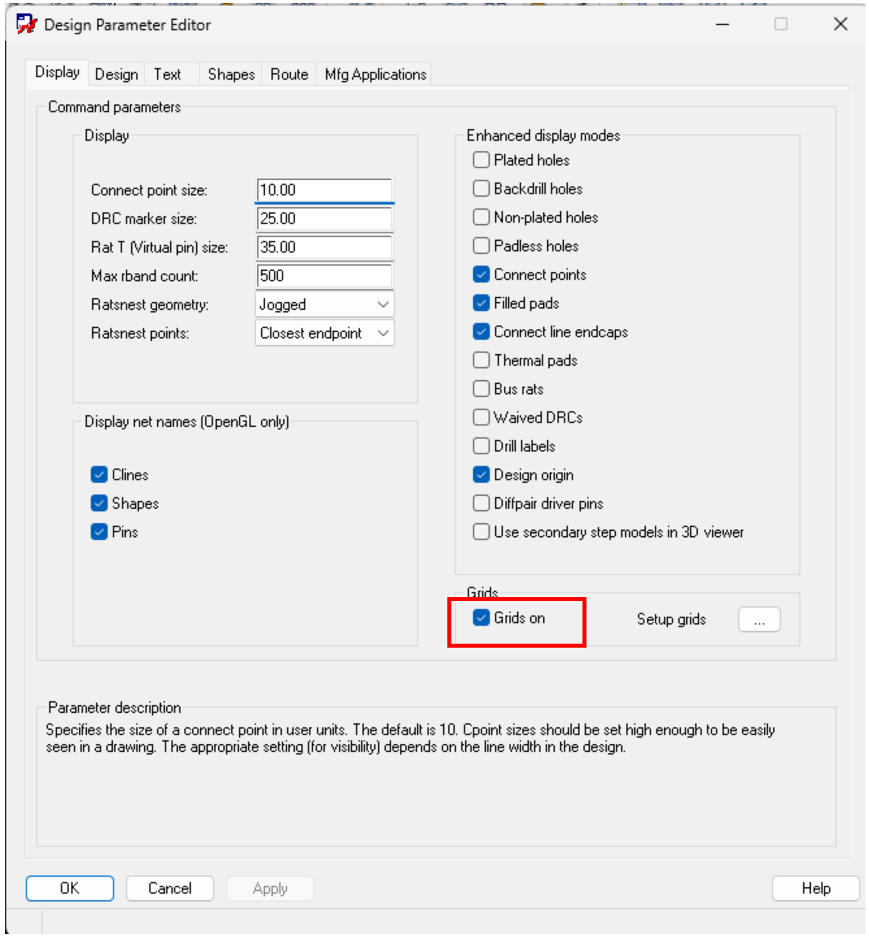
Create Netlist

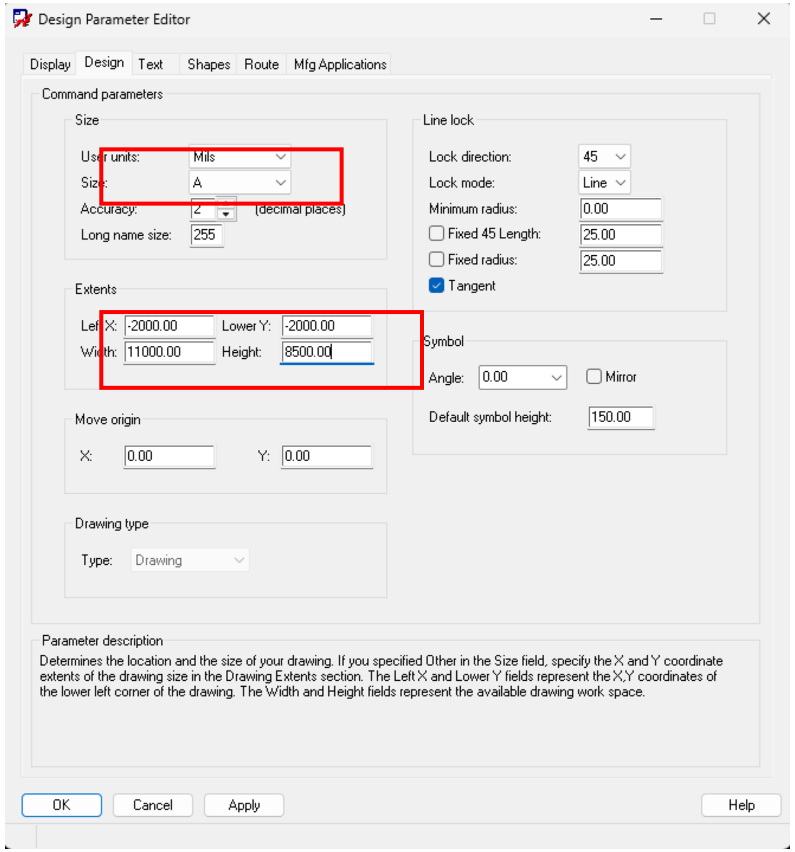


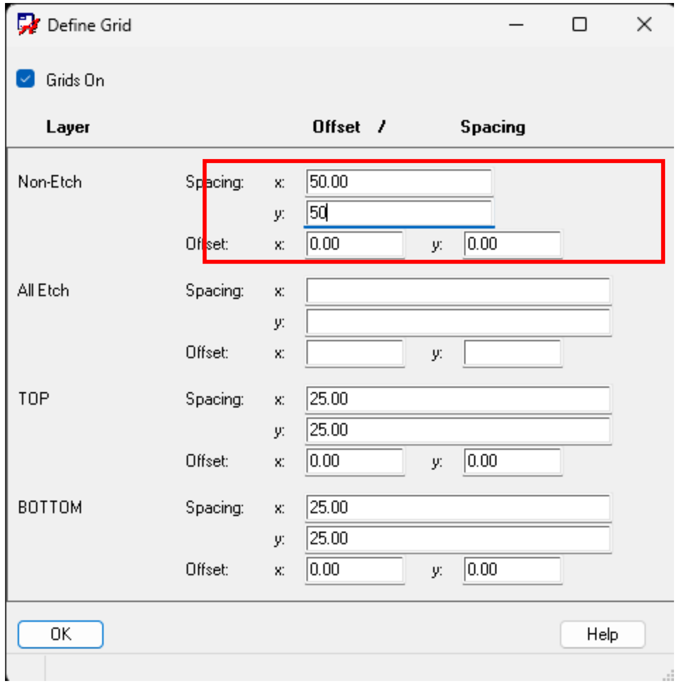


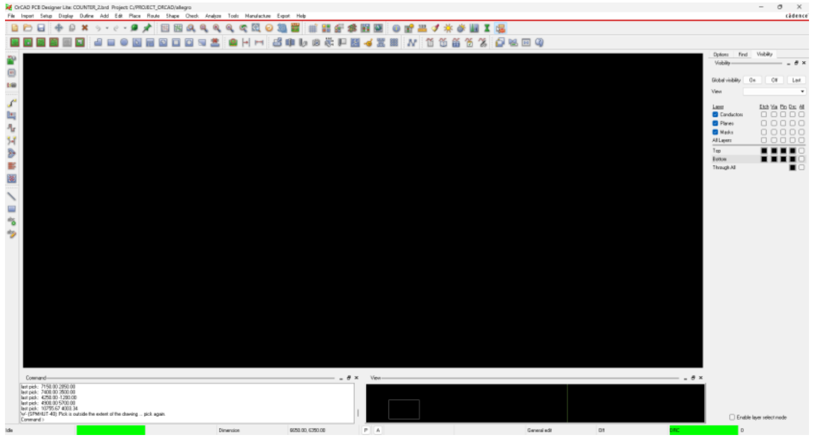
2. Pre-Layout Setup

PCB Editor Initial Settings

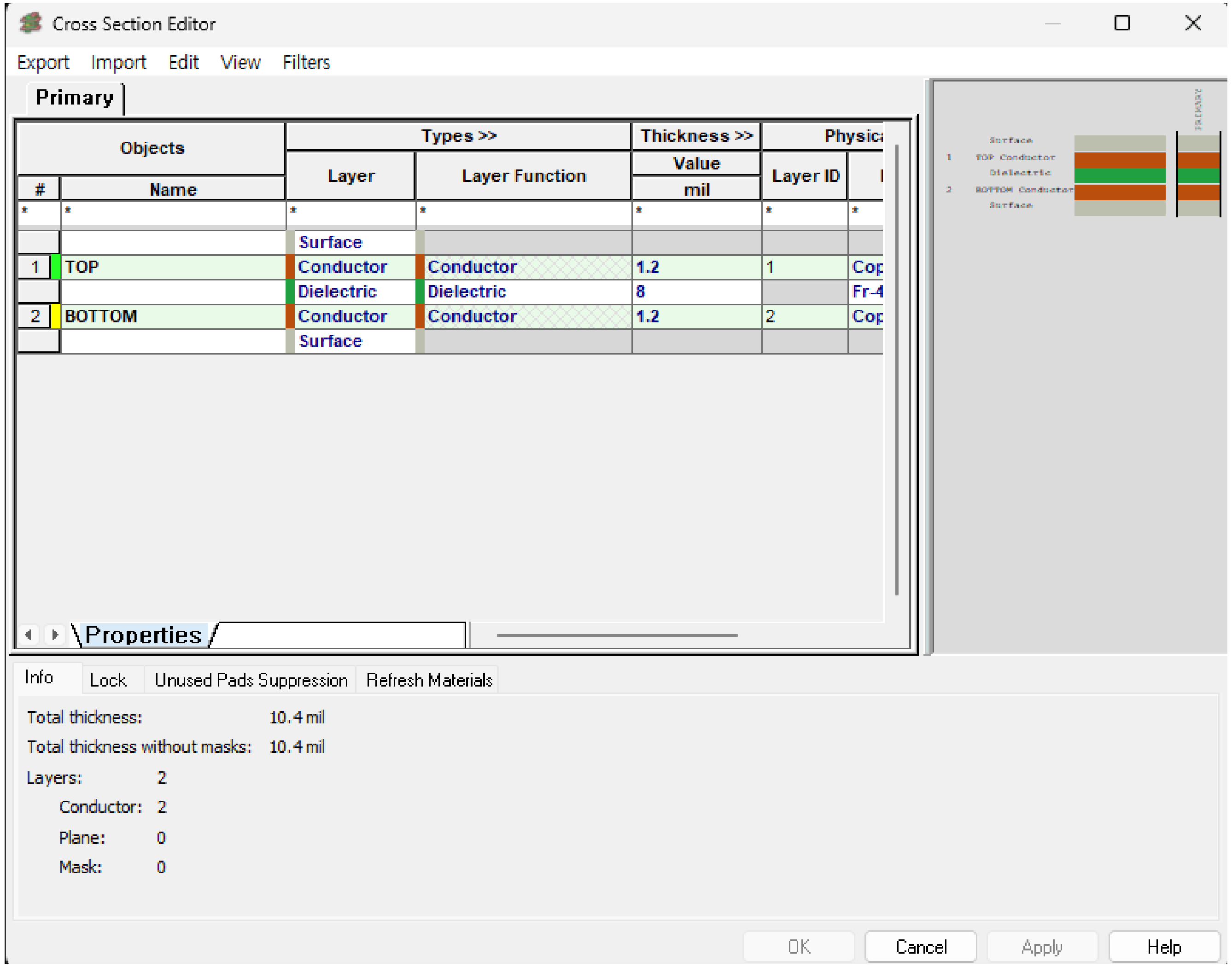






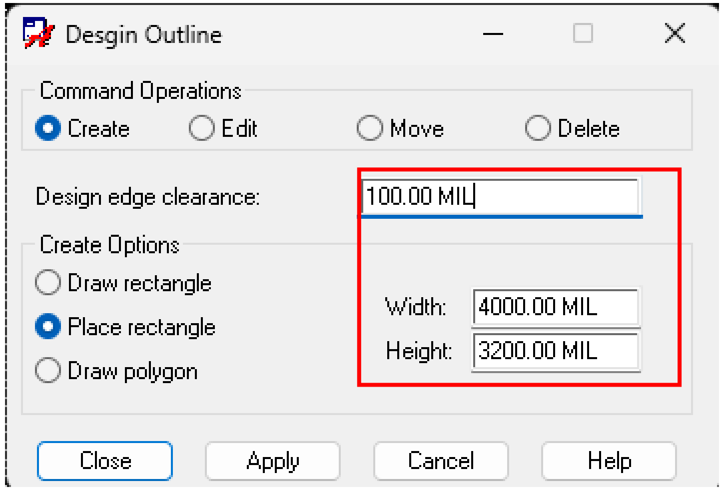


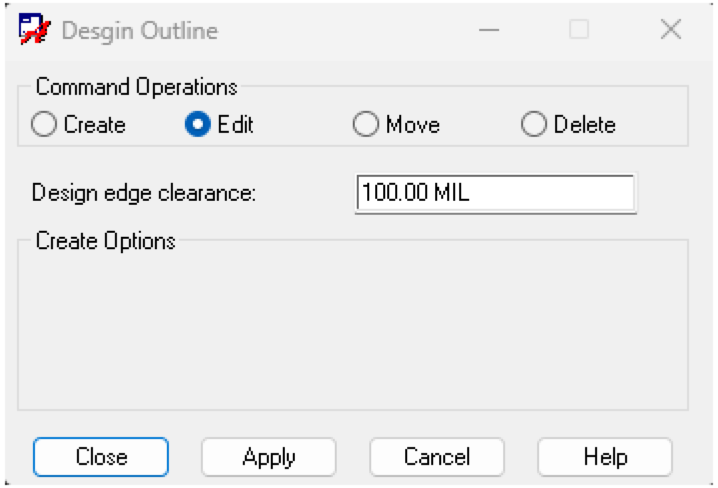
PCB Editor Initial Settings



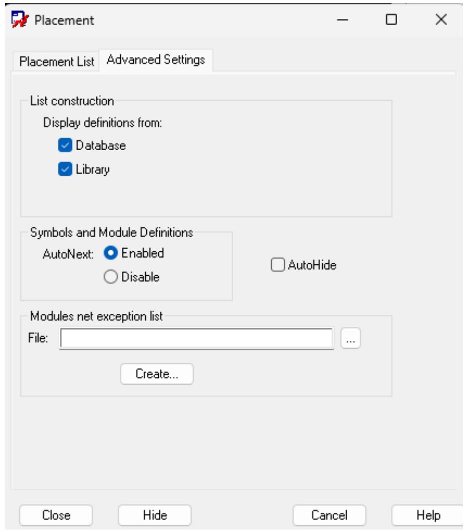
2-layer PCB

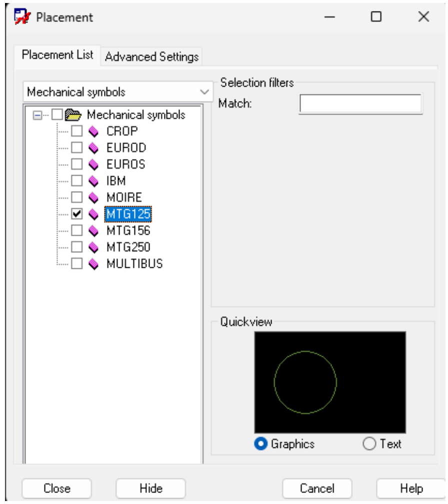
Board Outline Setup



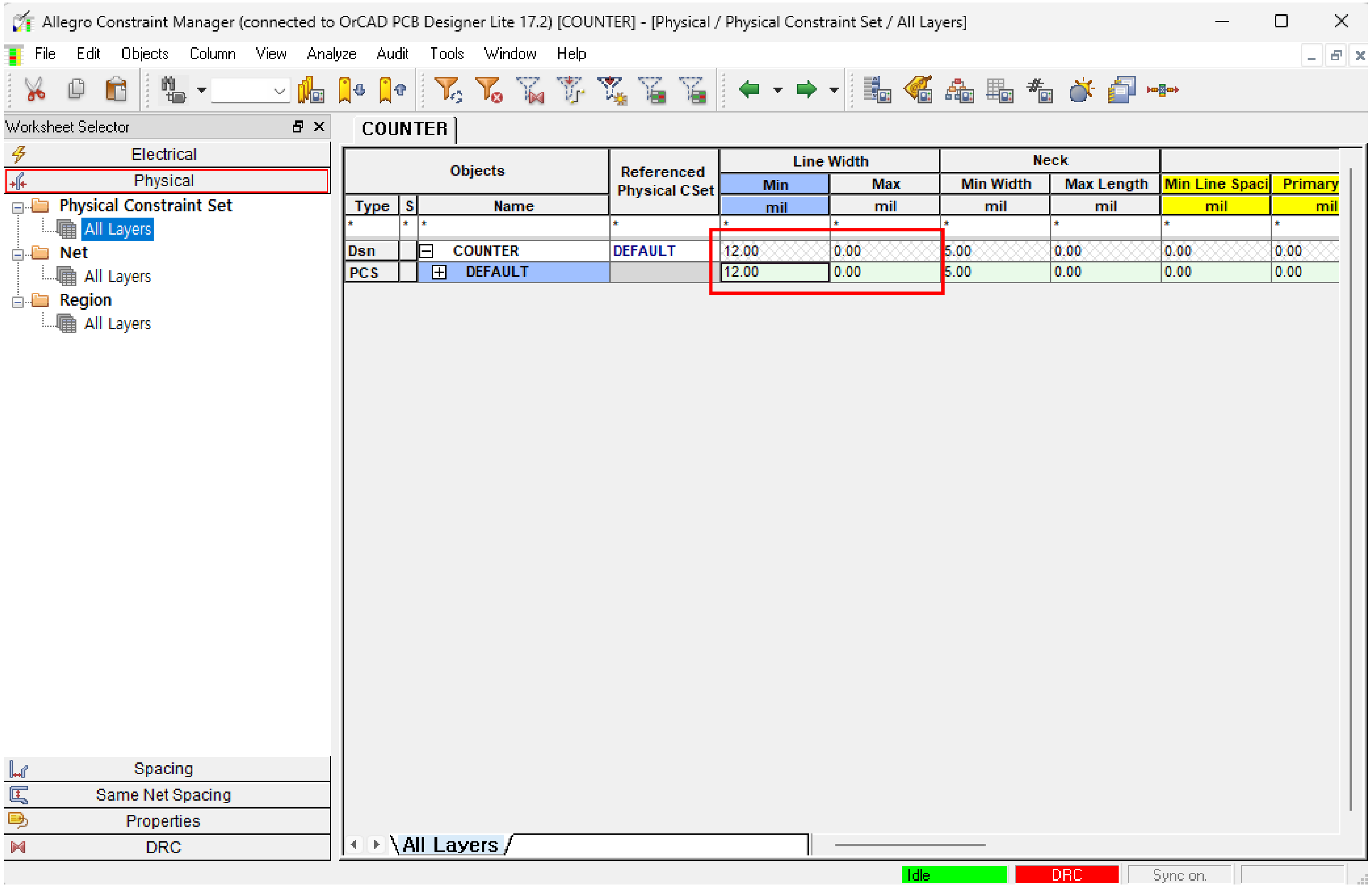


Board Outline Setup

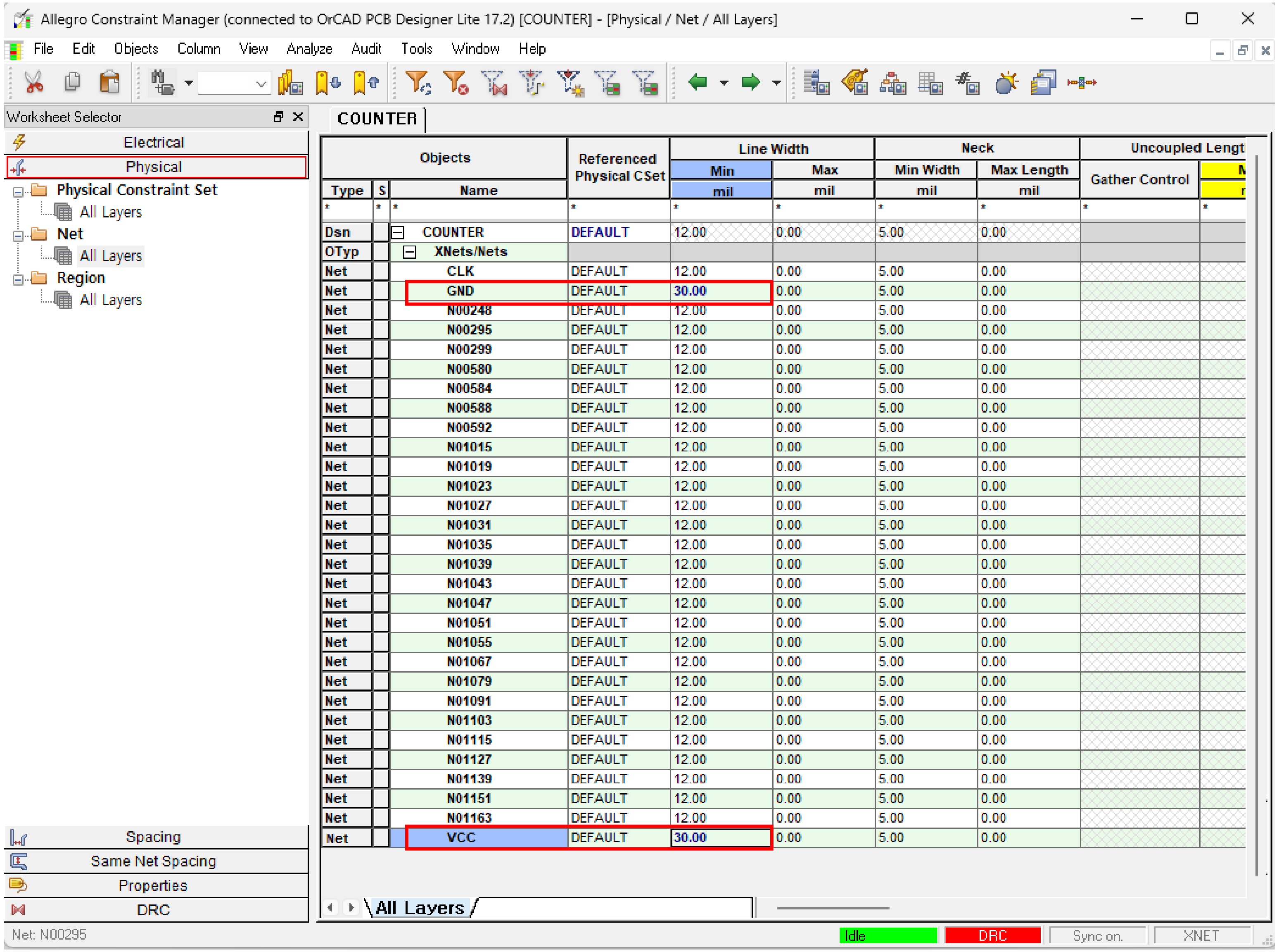




Constraints setup

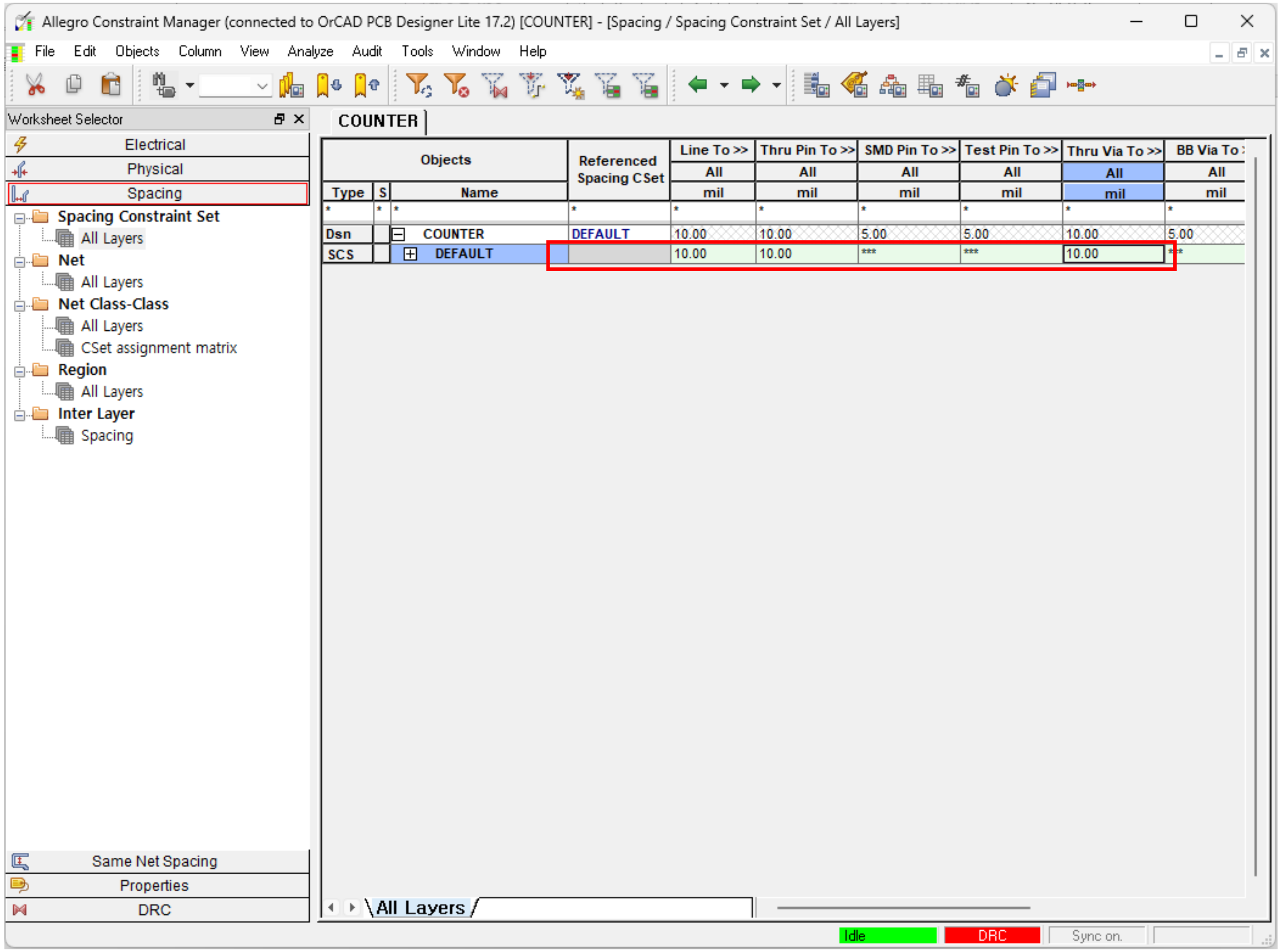


Constraints setup



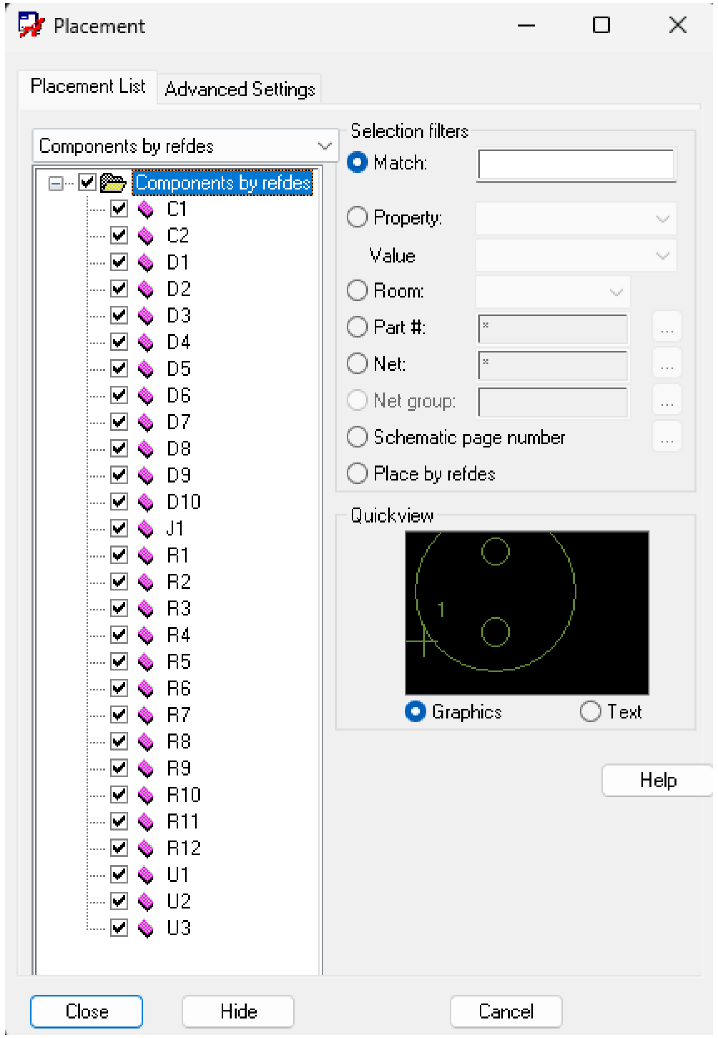
Type	S	Name	Referenced Physical CSet	Line Width Min (mil)	Line Width Max (mil)	Neck Min Width (mil)	Neck Max Length (mil)	Gather Control	Uncoupled Lengt
*	*	*	*	*	*	*	*	*	*
Dsn		COUNTER	DEFAULT	12.00	0.00	5.00	0.00		
OTyp		XNets/Nets							
Net		CLK	DEFAULT	12.00	0.00	5.00	0.00		
Net		GND	DEFAULT	30.00	0.00	5.00	0.00		
Net		N00248	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00295	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00299	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00580	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00584	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00588	DEFAULT	12.00	0.00	5.00	0.00		
Net		N00592	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01015	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01019	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01023	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01027	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01031	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01035	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01039	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01043	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01047	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01051	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01055	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01067	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01079	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01091	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01103	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01115	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01127	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01139	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01151	DEFAULT	12.00	0.00	5.00	0.00		
Net		N01163	DEFAULT	12.00	0.00	5.00	0.00		
Net		VCC	DEFAULT	30.00	0.00	5.00	0.00		

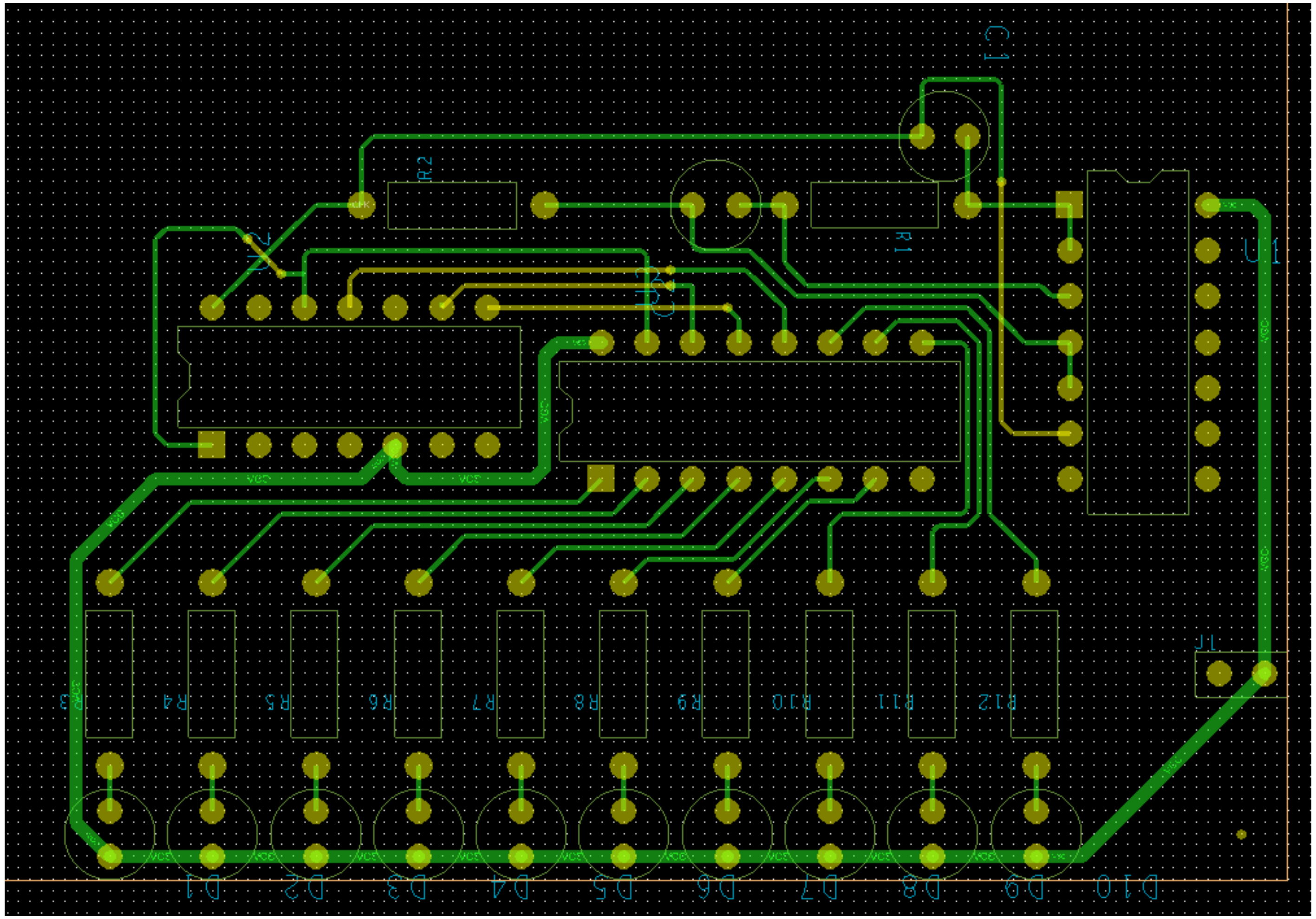
Constraints setup



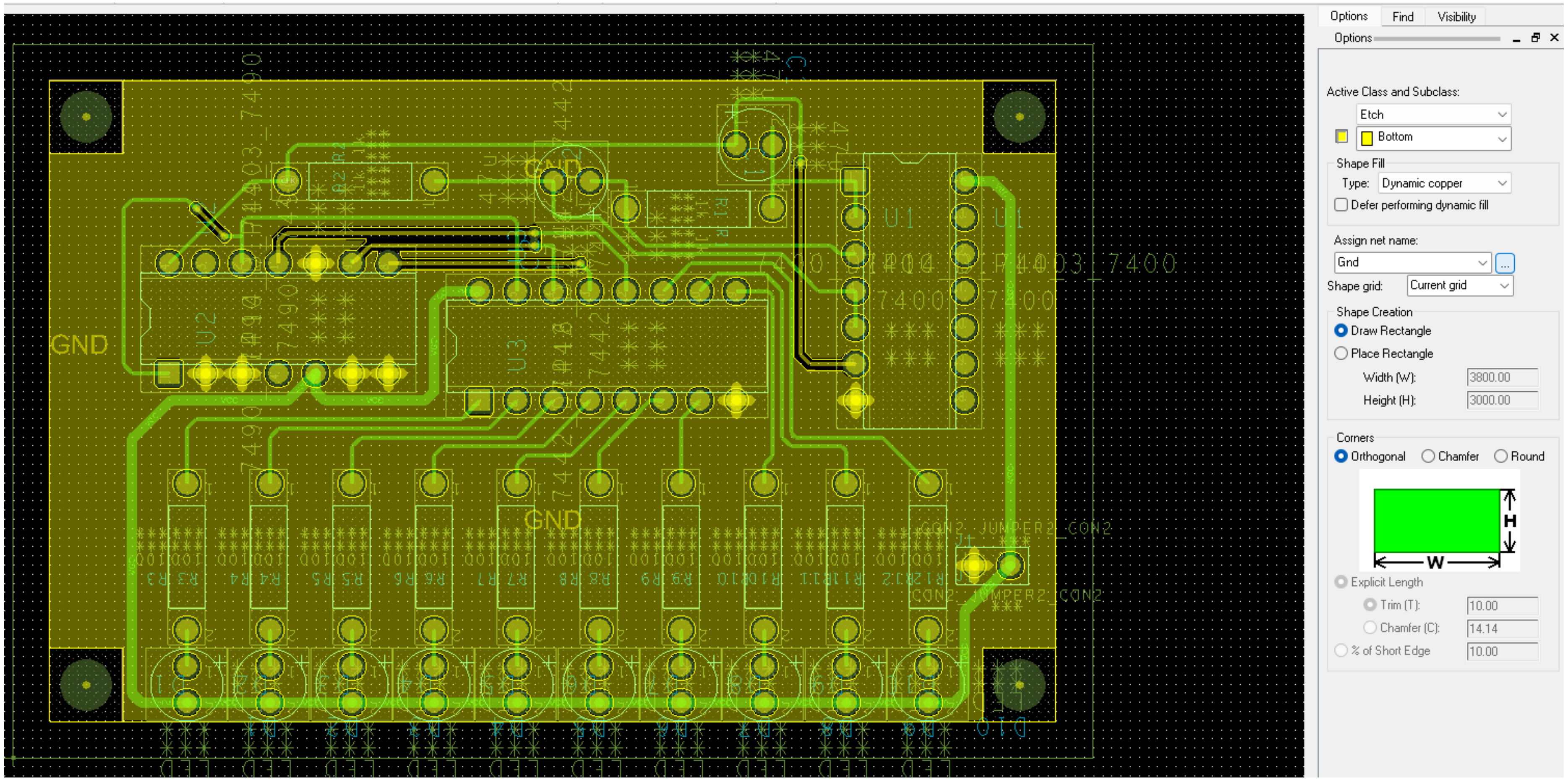
3. Layout

Part Placement and Routing

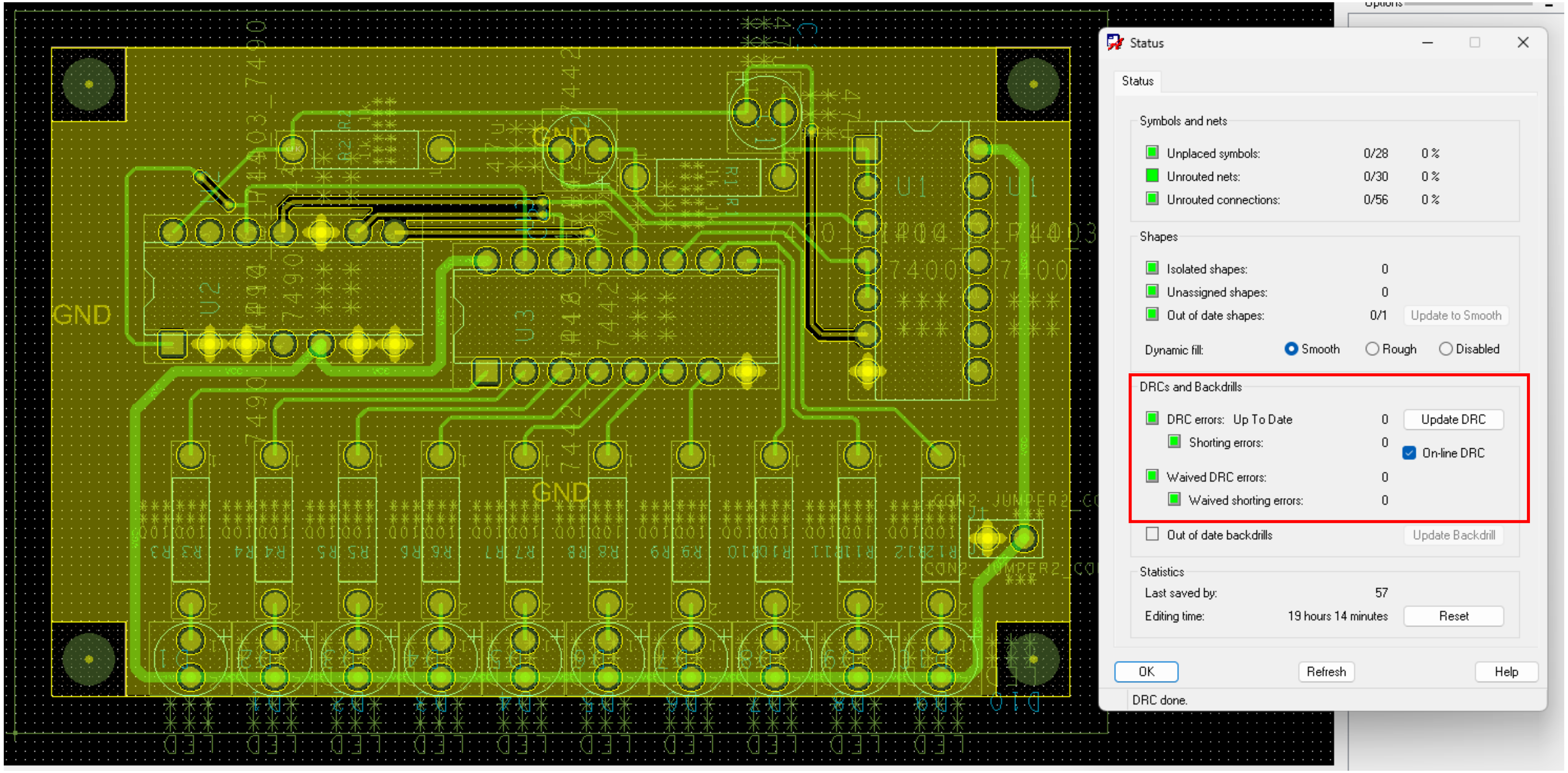




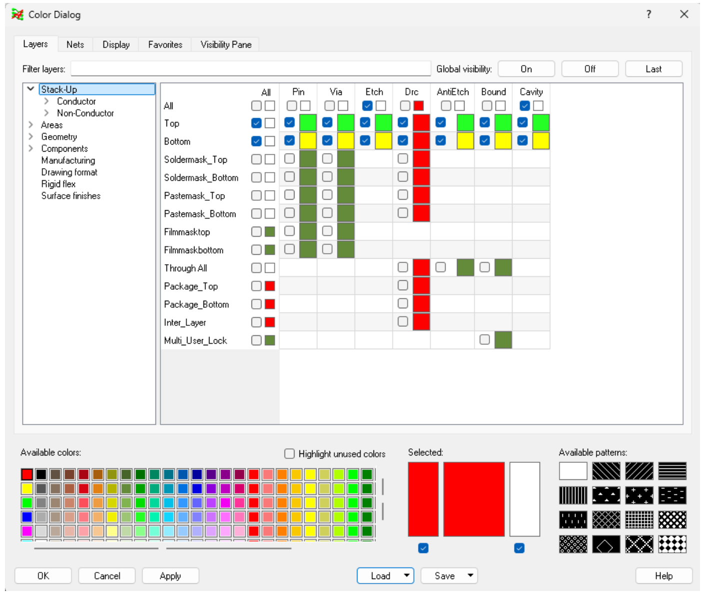
Ground Copper Pour Placement

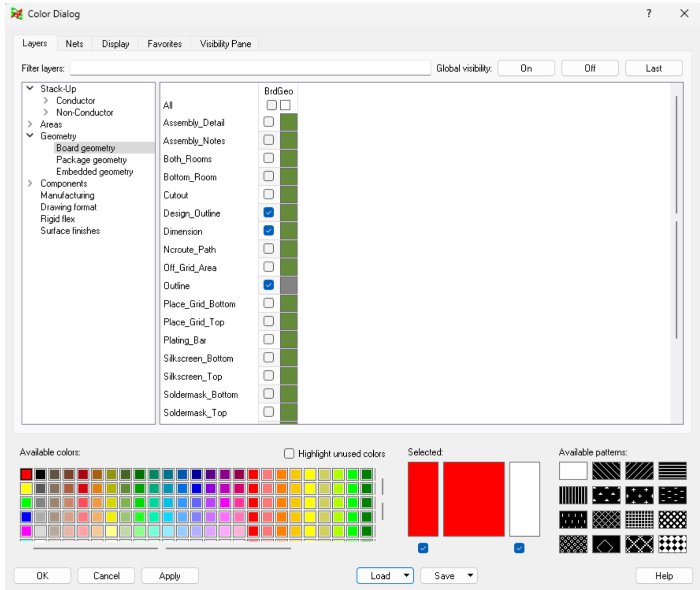


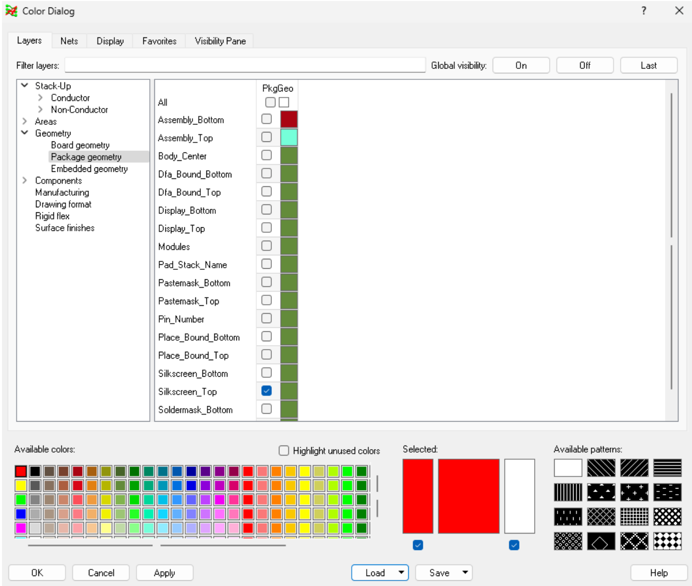
Design Rule Check (DRC)

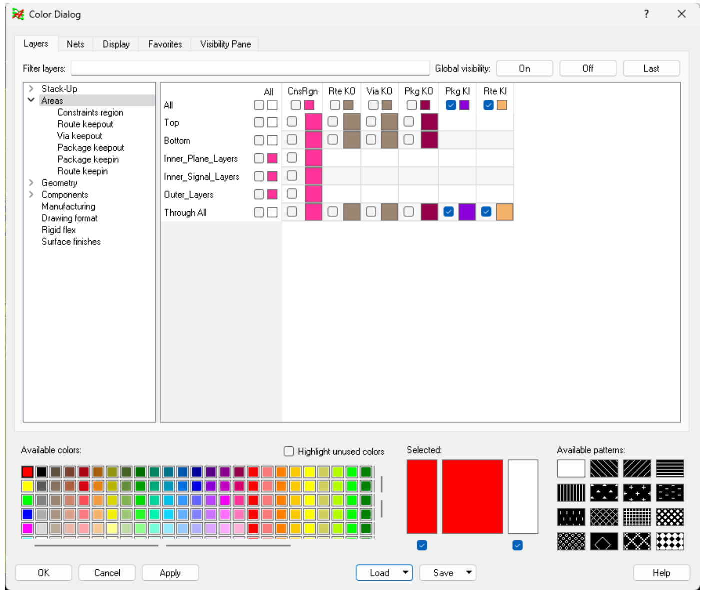


Color Setting

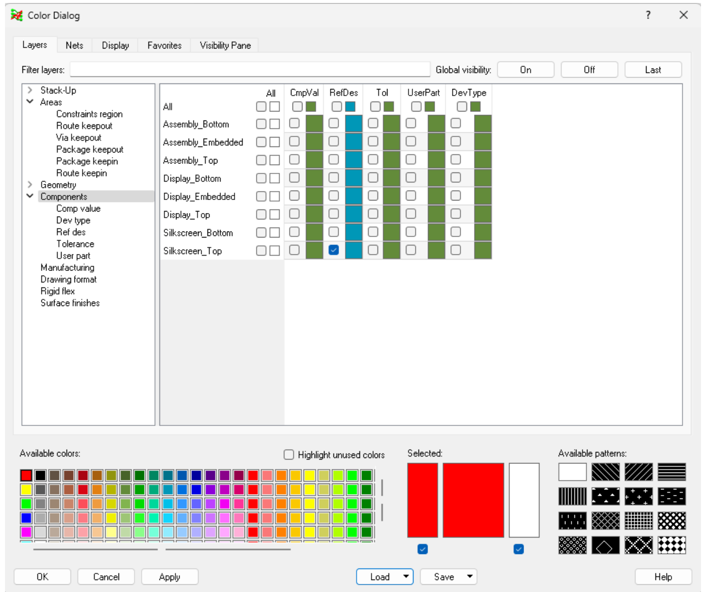


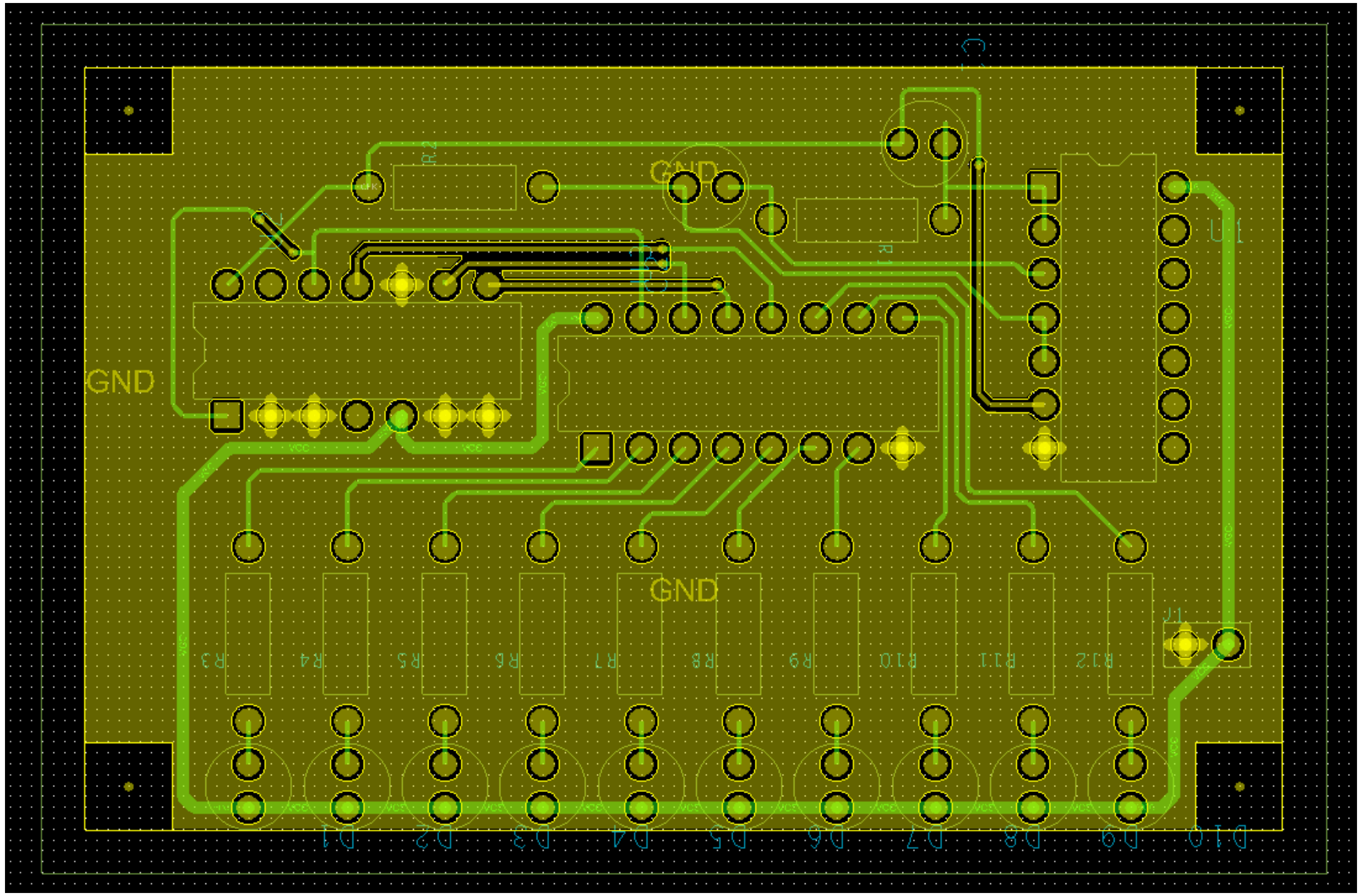




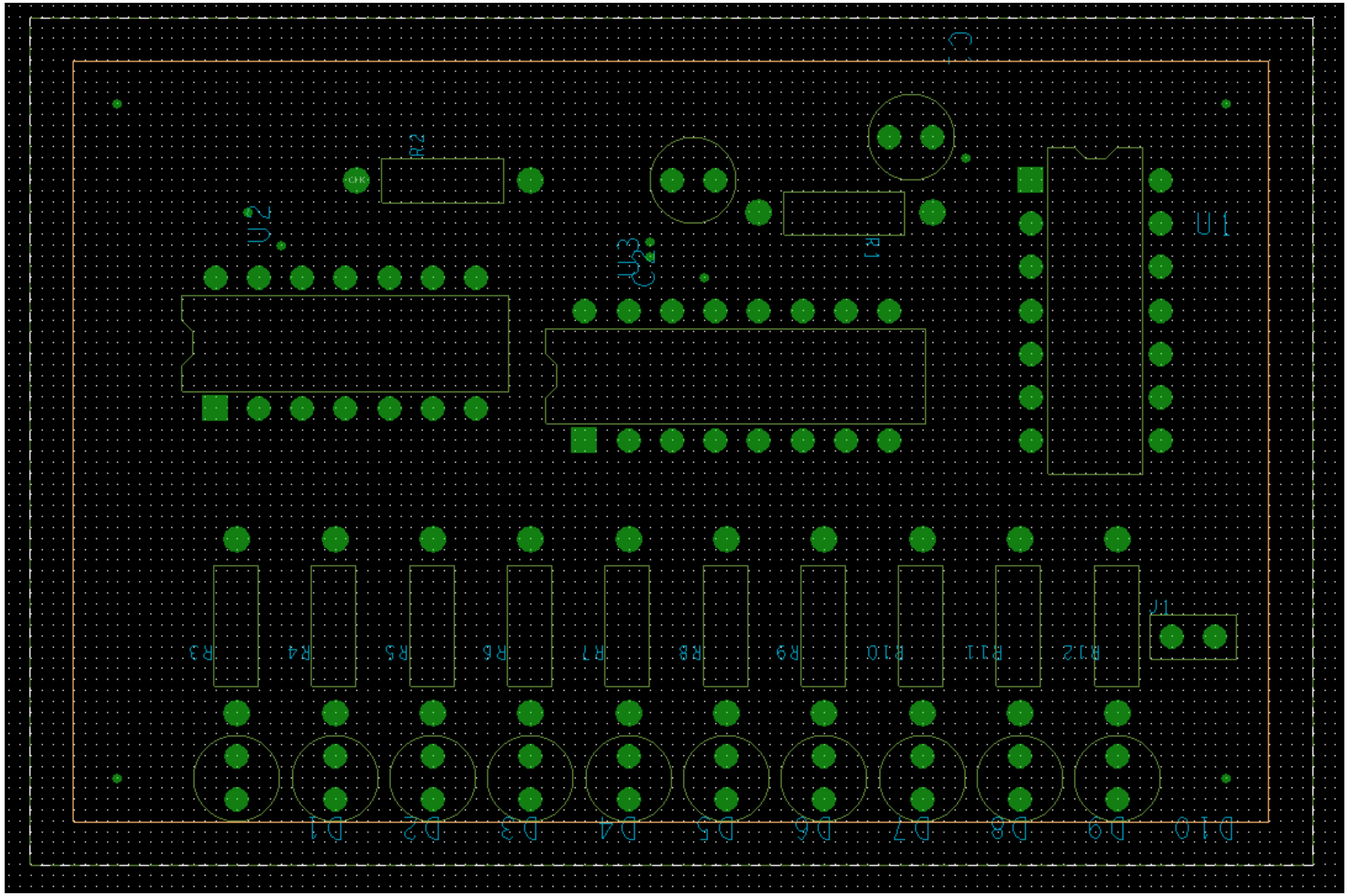


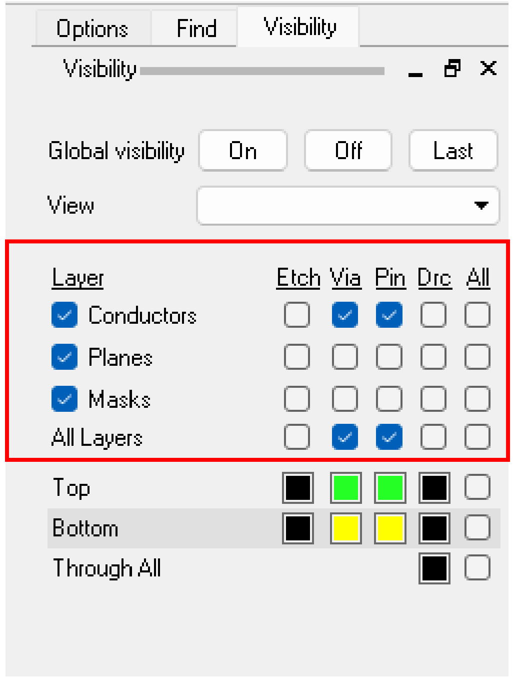
Color Setting





Reference Annotation





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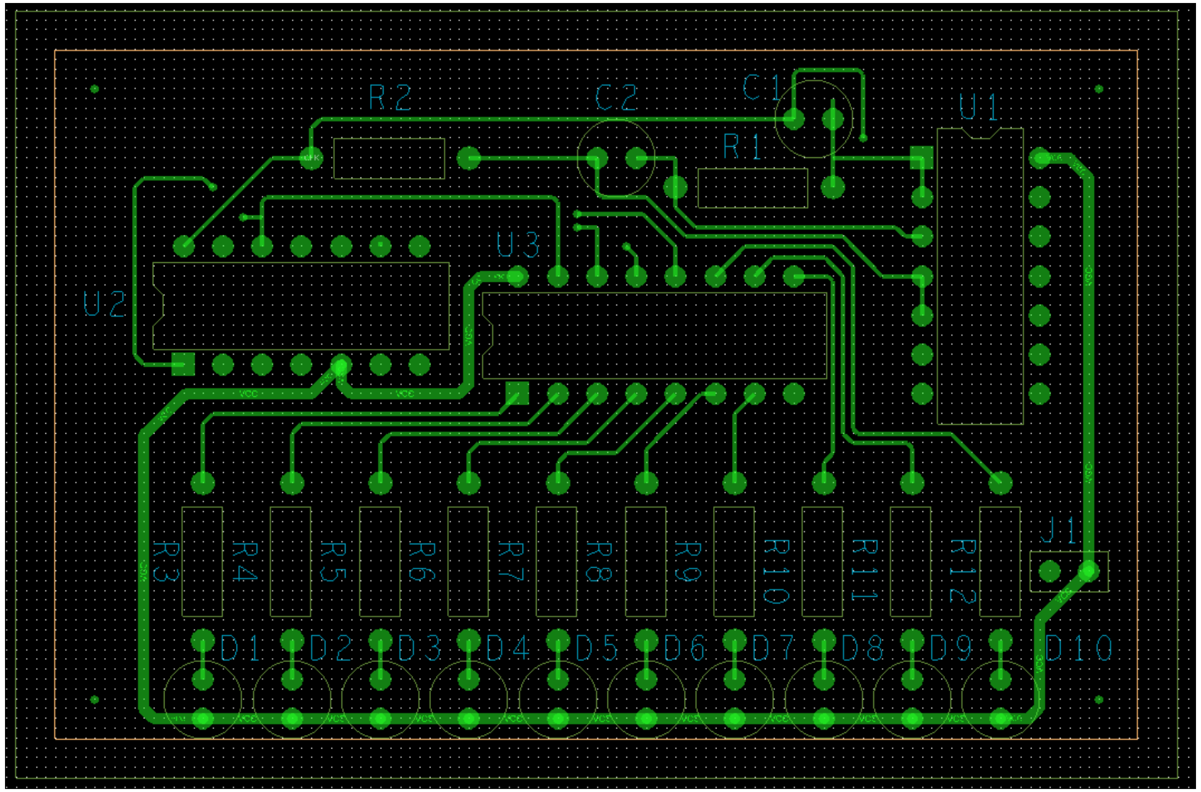
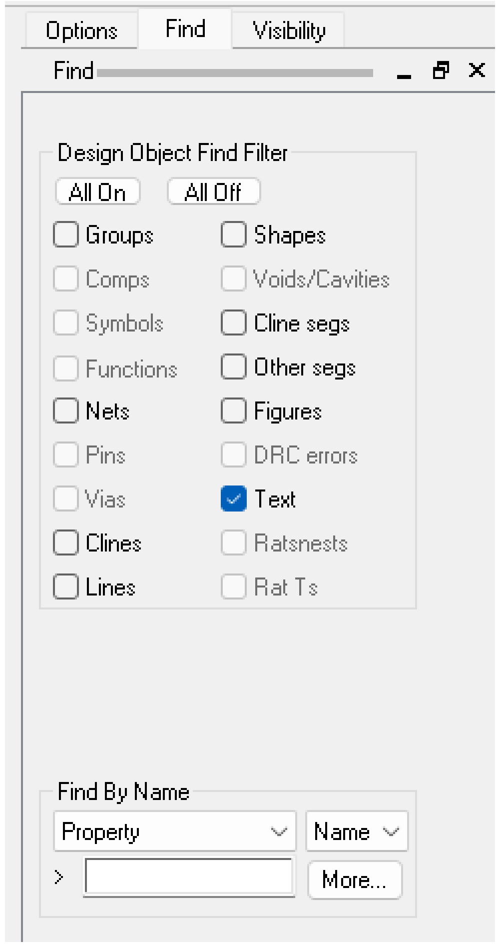
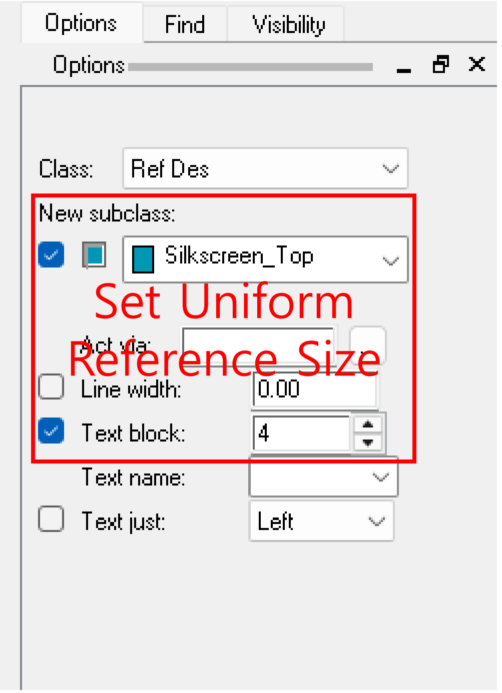
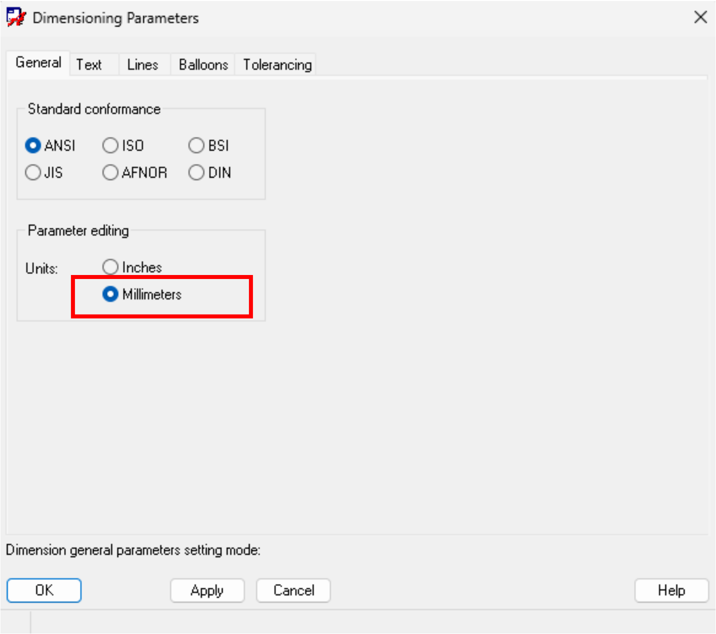


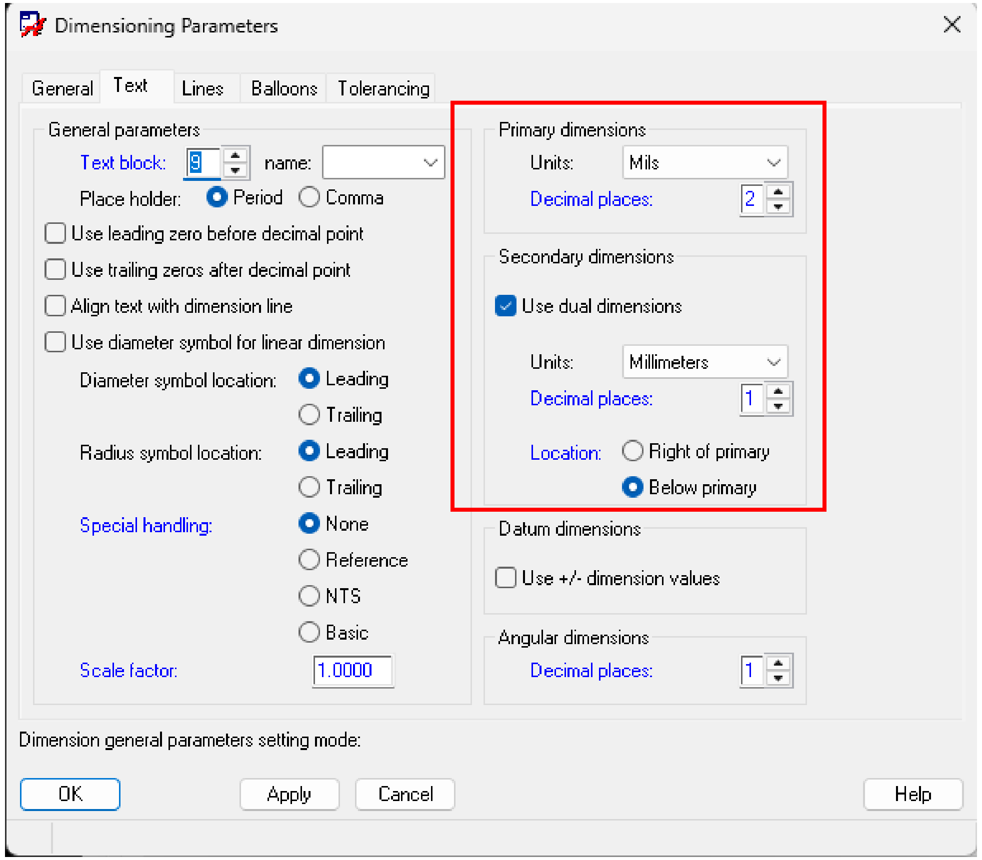




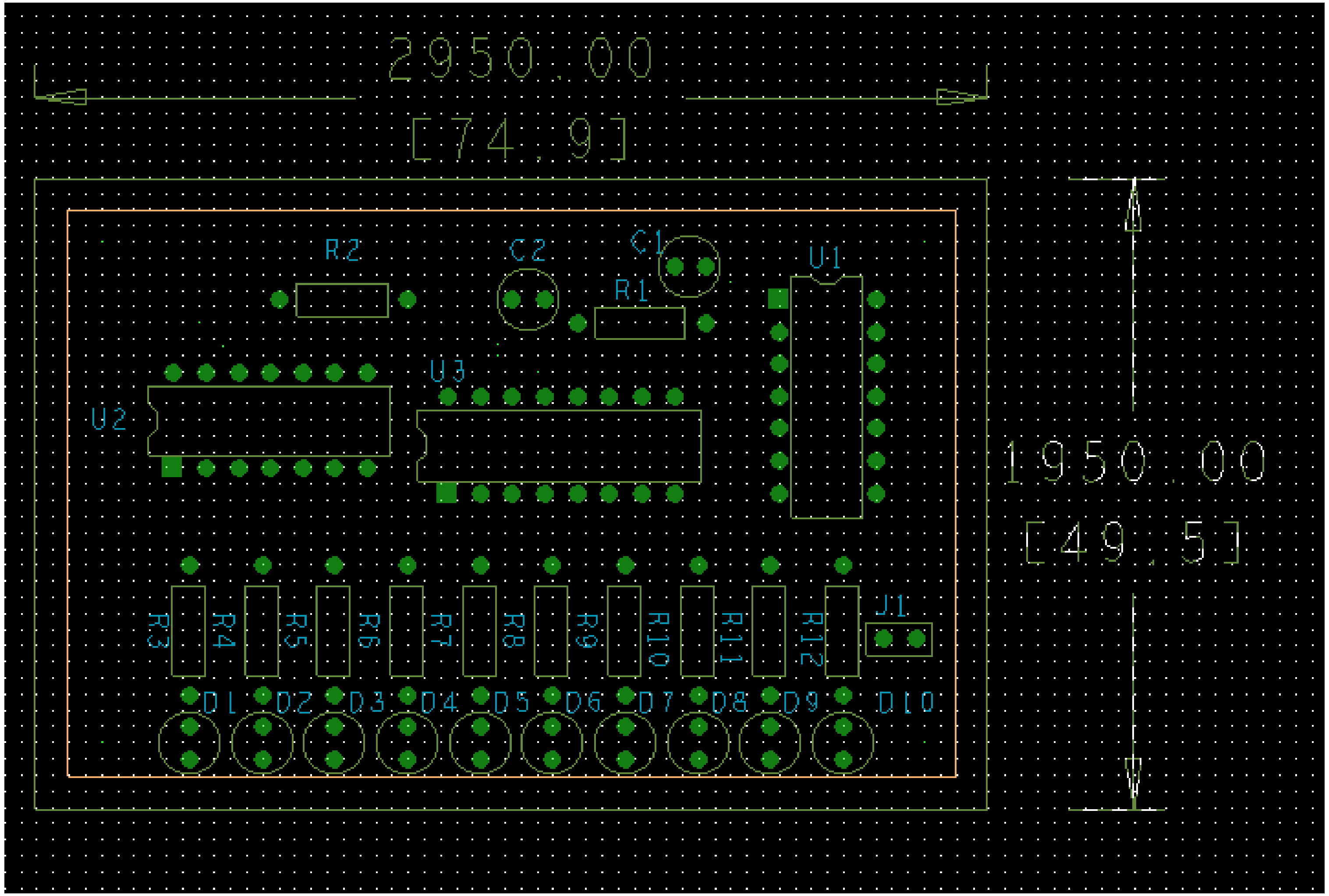
4. Drafting & Dimensions

Drafting Dimension Settings

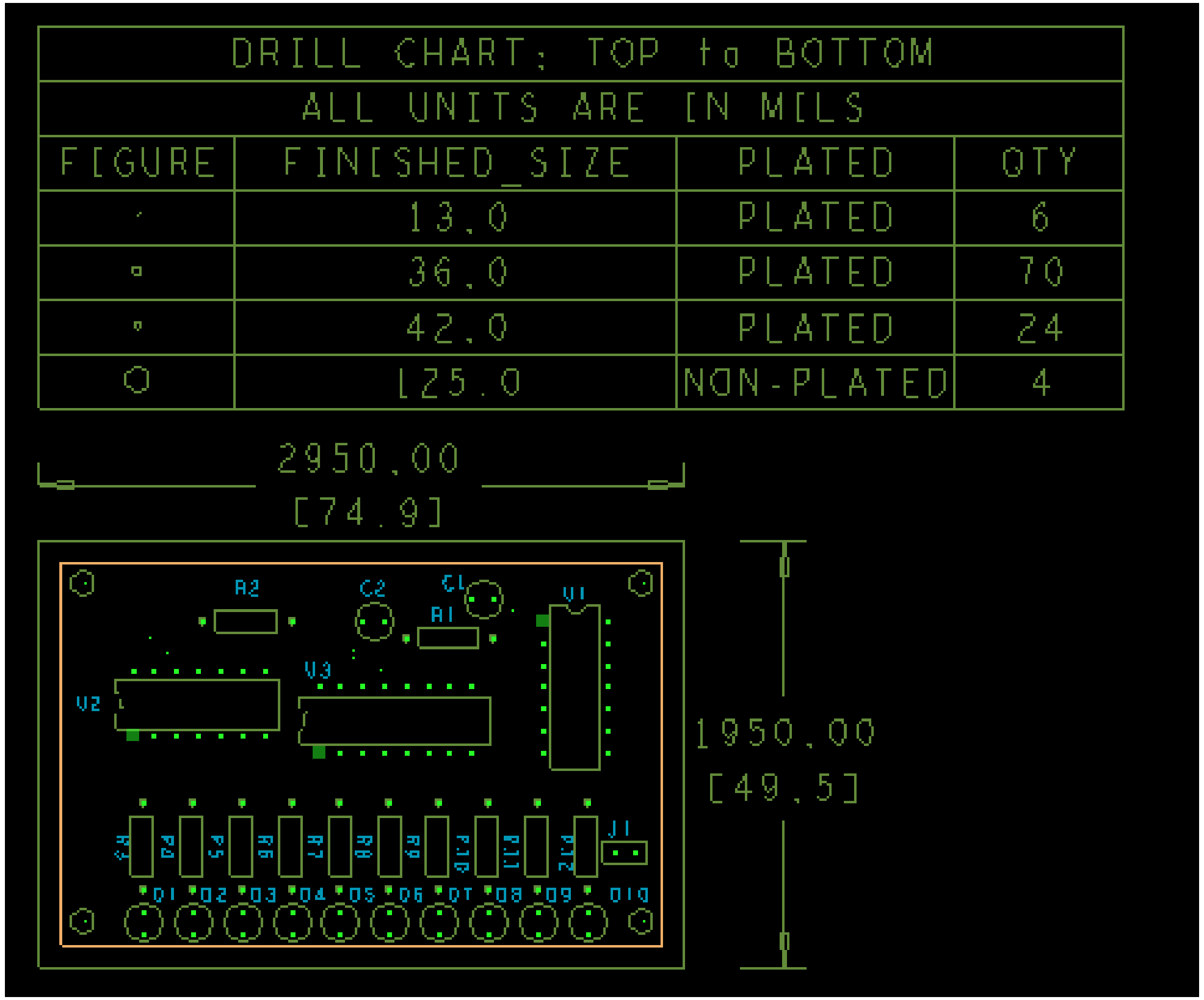


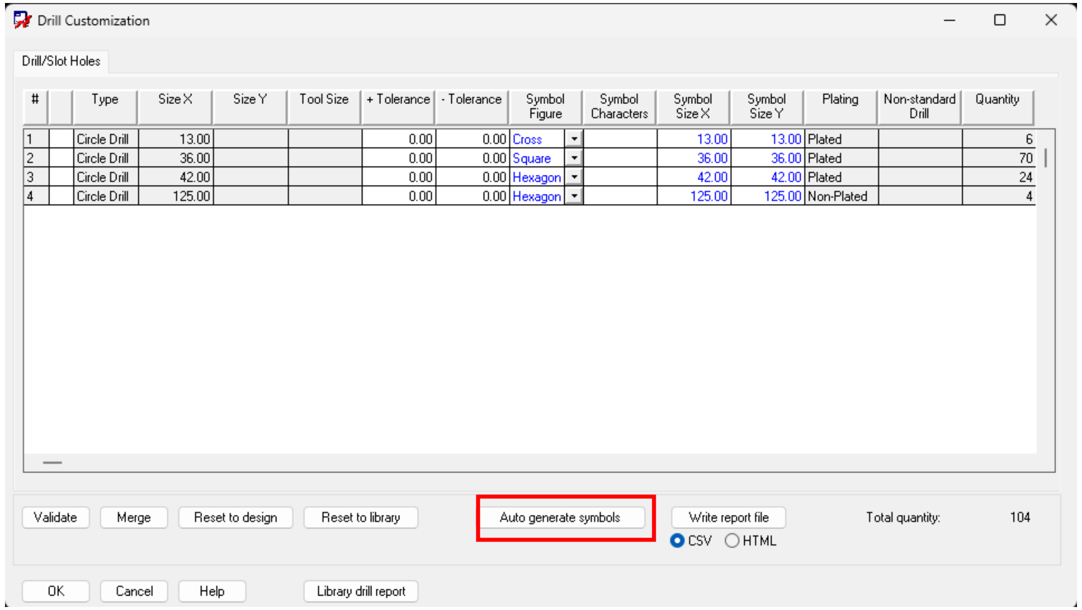


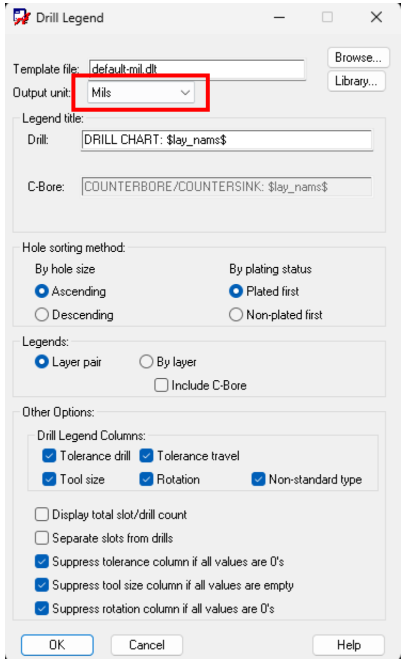
Add Dimensions

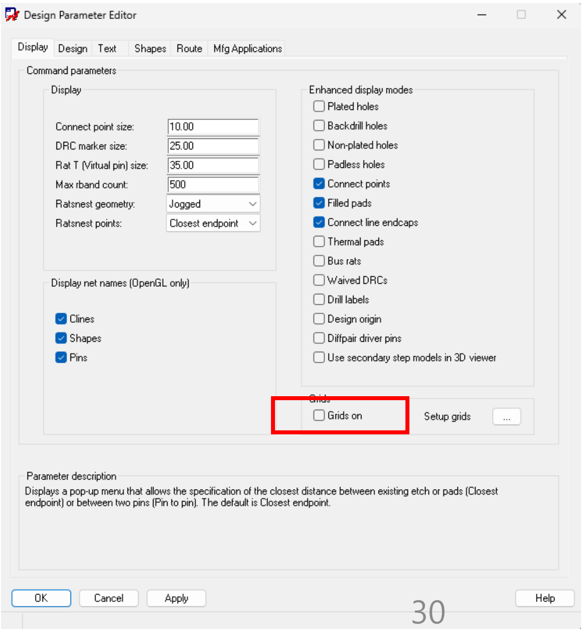


Add Drill Legend Table



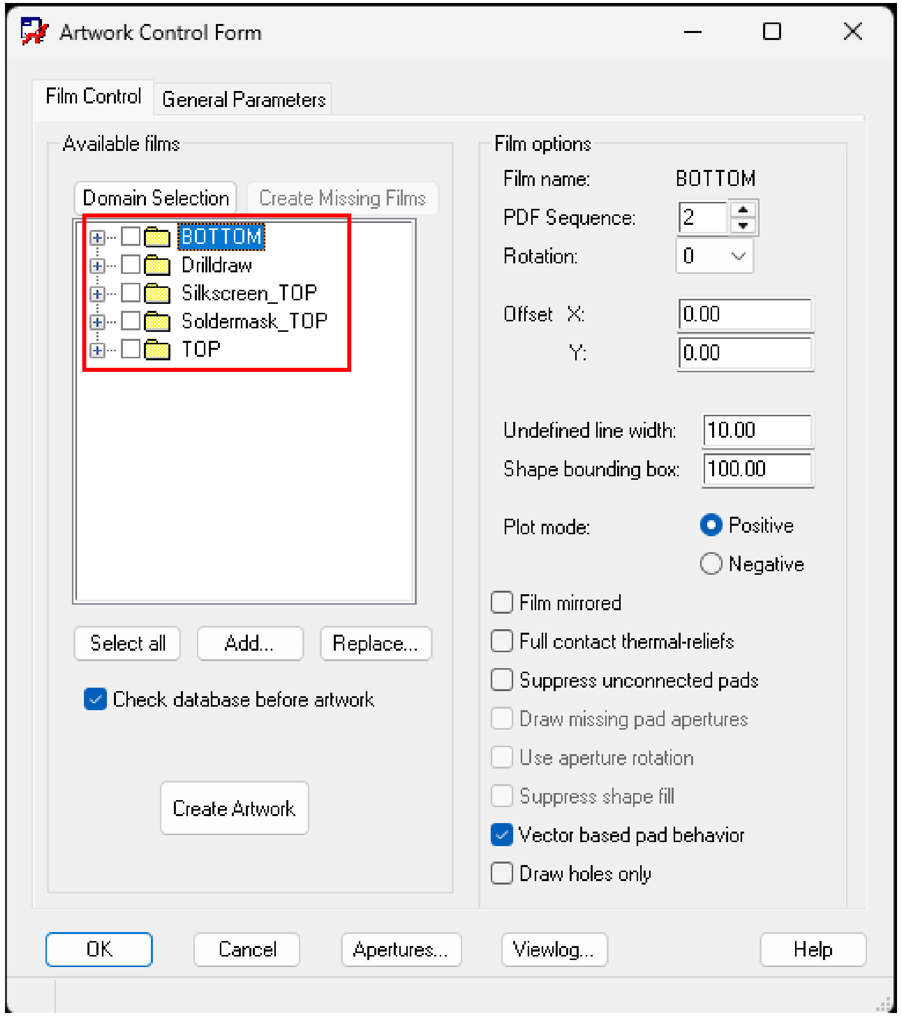






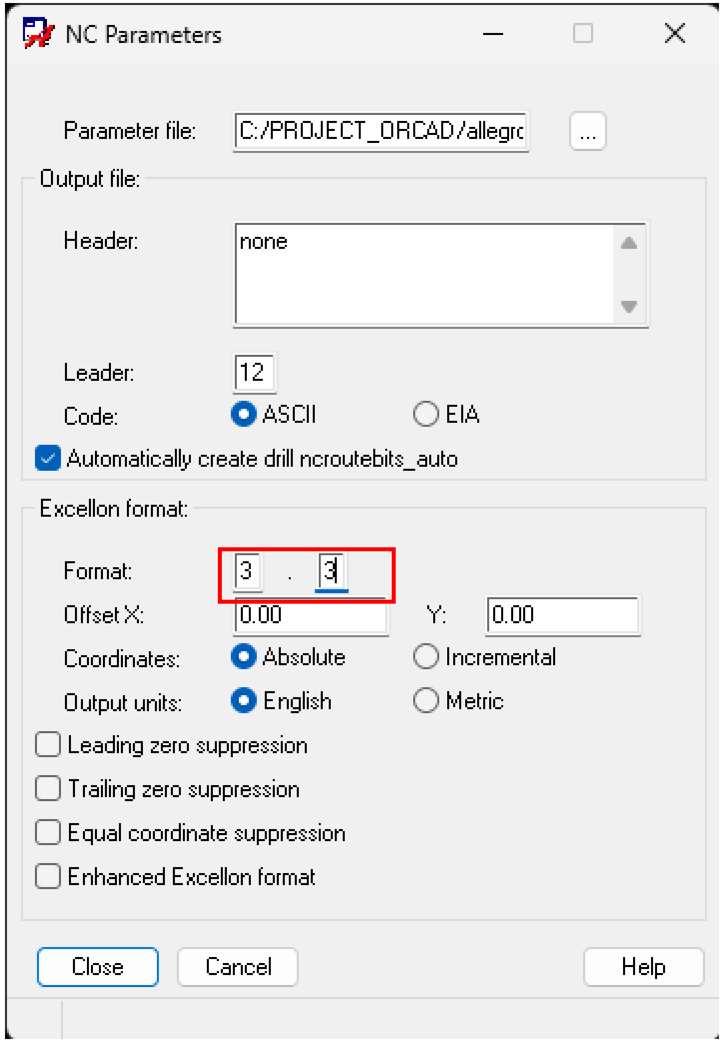
5. Manufacturing Outputs

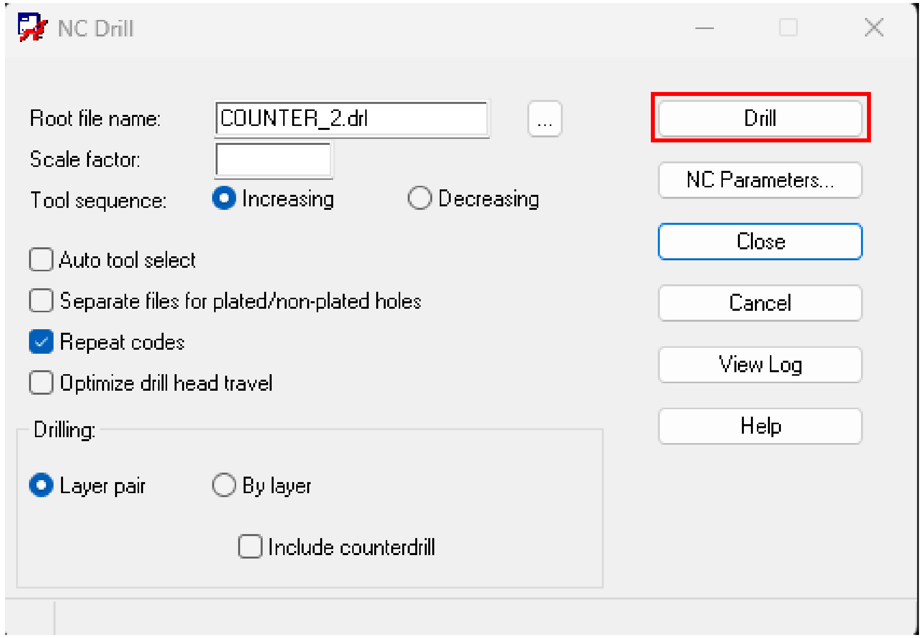
Generate Gerber Files



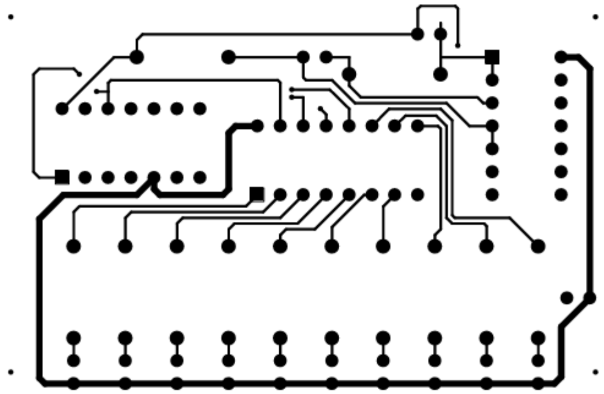
Generate Gerber Files
- TOP, BOTTOM, Soldermask_TOP, Silkscreen_TOP, Drilldraw

Generate NC Drill

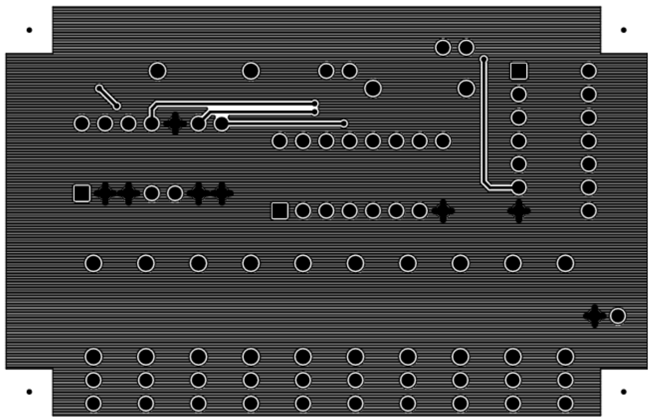




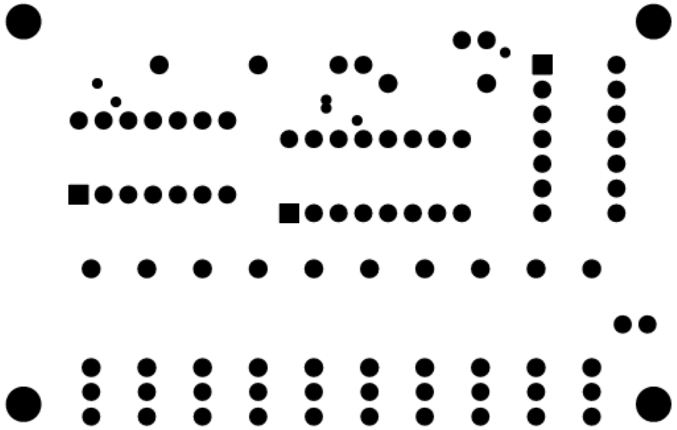
PCB Gerber Files



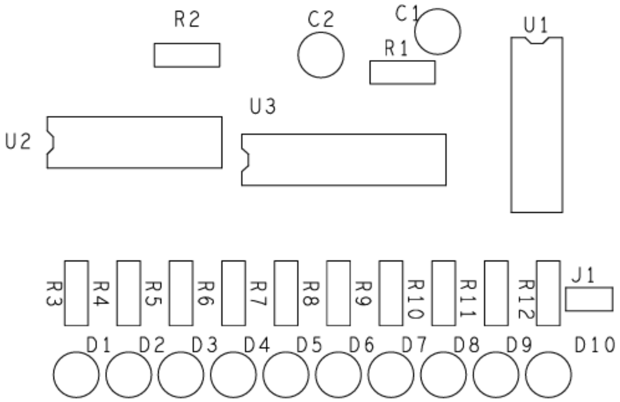
<Top Layer Pattern>



<Bottom Copper Layer>



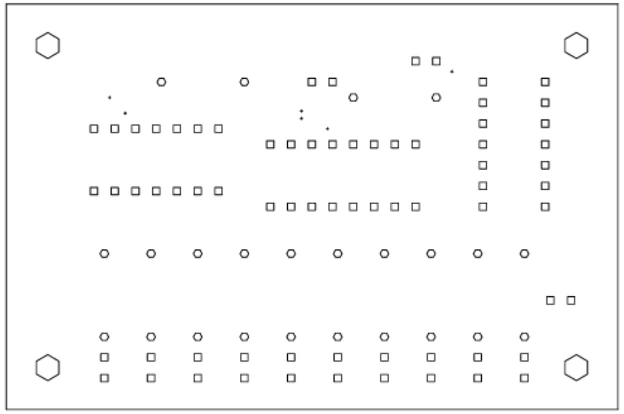
<Top Silkscreen>



<Top Soldermask>

PCB Gerber Files

DRILL CHART: TOP to BOTTOM			
ALL UNITS ARE IN MILS			
FIGURE	FINISHED_SIZE	PLATED	QTY
·	13.0	PLATED	6
□	36.0	PLATED	70
○	42.0	PLATED	24
⬡	125.0	NON-PLATED	4



<Drilldraw>

CONTACT

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emh0412@naver.com

Mobile

+82 10 6616 2655